Chromosphere2026 Technical Record

Physics, Numerical Methods, Implementation, Validation, and Development History

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Release baseline documented: August 2026

Abstract

This document is the engineering and numerical record for Chromosphere2026. It describes the current `model_column` release, the physical and numerical models from which it evolved, the validation evidence supporting the implementation, the approaches that were rejected or retired, and the limitations that remain. It is not a journal article and it is not a chronological change log.

The active release is a one-dimensional field-aligned hydrogen hydrodynamic column with a common velocity and temperature, equilibrium Saha ionization, conserved hydrogen ionization energy, and a density- and temperature-dependent acoustic exponent Γ_1 supplied by a validated CRASH-derived table. The atmosphere is initialized from the Model C7 temperature profile using shape-preserving cubic interpolation and is reconstructed in Saha-consistent hydrostatic equilibrium. Hydrodynamics uses a nonuniform finite-volume MUSCL–Hancock method that reconstructs $(\log \rho,\ v,\ \log p)$ and advances an SWMF-style exact-Riemann Godunov flux evaluated at frozen composition. It is reference-free: no frozen hydrostatic state is stored or subtracted. Thermal conduction is nonlinear and implicit. Static local refinement resolves the upper transition-region structure; the mesh-scaled artificial conductivity is implemented but disabled in the release configuration, which uses physical conduction only. The validated runtime supports deterministic OpenMP execution.

The repository also contains two-fluid, kinetic-ionization, radiative, flare-heating, three-temperature, loop, and multidimensional formulations. These remain technically useful, but they are not all part of the active `model_column` release. Status labels are used throughout to prevent historical or optional material from being mistaken for current production behavior.

Contents

Status Legend and Reading Guide

Label	Meaning
Active release path	Used by the validated gamma-table `model_column` release configuration described in this document.
Implemented but inactive	Present in the code and testable, but disabled in the reference release calculation.
Experimental	Implemented for controlled studies; validation or physical interpretation is not sufficient for release use.
Historical	Important to the design history or still used as a regression baseline, but superseded as the primary model.
Retired	Deliberately removed from the active path or rejected after testing.
Planned	A documented future capability that is not yet implemented or validated.

Two configurations must be distinguished. The *software defaults* are conservative fallbacks used when environment variables are absent. The *validated release configuration* is the specific 1600–2153 km, outer-refined, gamma-table calculation used for the current convergence, performance, and long-run evidence. A claim about one does not automatically apply to the other.

Part I

Project Overview and Release Baseline

Chapter 1

Purpose and Scope

1.1 Scientific motivation

The chromosphere and transition region mediate mass and energy exchange between the dense lower atmosphere and the corona. Conductive energy arriving from above can be distributed among sensible heating, hydrogen ionization, radiation, mechanical work, and upward enthalpy transport. A lower boundary that fixes thermodynamic variables cannot represent this response dynamically. Chromosphere2026 therefore develops a field-aligned column that can evolve the lower atmosphere and eventually provide time-dependent mass and energy fluxes to a global coronal model.

The project is designed as a future lower-atmosphere component for AWSoM within SWMF. **Planned** The present repository does not yet perform that coupling. Its current role is to establish a thermodynamically consistent and numerically controlled column model whose boundary fluxes can later be exposed to a global calculation.

The initial atmospheric data are based on Model C7 of Avrett and Loeser [2008]. The broader two-fluid lineage follows chromospheric plasma-neutral formulations such as Al Shidi et al. [2019], while the current release deliberately takes the strong-coupling equilibrium limit rather than retaining independent ion and neutral dynamics.

1.2 Software scope

The repository contains three overlapping layers:

1. a separate, non-release seven-row plasma-neutral state representation and source-stage framework;
2. several scenarios that exercise different physical closures and geometries; and
3. the current `model_column` release path, a separate single-fluid solver that evolves the three-variable equilibrium mixture directly.

The two capabilities are now two separate solvers over one shared mesh. The release solver advances the three-variable single-fluid state $(\rho, \rho v, E)$ and never touches the seven-row representation; the seven-row representation belongs exclusively to the historical two-fluid/multi-temperature/finite-rate-ionization research path. Selecting a Γ_1 table selects the release solver, and each solver rejects the other's Grid outright.

Chapter 2

Exact Release Baseline

2.1 Active calculation

Active release path

The release baseline consists of the following coupled choices:

- one-dimensional field-aligned hydrodynamics on a prescribed straight field line;
- constant solar gravity along the vertical column;
- pure hydrogen with a common bulk velocity and common temperature, advanced as the three conserved variables $(\rho, \rho v, E)$;
- equilibrium ionization from the Saha relation;
- pressure and internal energy evaluated consistently from that equilibrium composition;
- explicit inclusion of hydrogen ionization energy in the conserved mixture energy;
- caloric inversion $e_{\text{int}}(\rho, T) \mapsto T$ and tabulated $\Gamma_1(\rho, T)$;
- Model C7 temperature initialization with PCHIP interpolation;
- Saha-consistent hydrostatic pressure and density reintegration;
- finite-volume MUSCL–Hancock time stepping with log-space reconstruction of $(\log \rho,\ v,\ \log p)$ and the MC3/Koren limiter, the predictor carrying the same momentum source as the corrector;
- the SWMF-style exact-Riemann Godunov numerical flux at frozen composition ($\gamma = 5/3$, ionization energy carried in a passive offset), with face-local Rusanov fallback;
- reference-free hydrostatic balance (no frozen equilibrium state);
- nonlinear backward-Euler physical thermal conduction;
- static local refinement near the upper transition region;
- no artificial or numerical conduction anywhere in the production operator;
- no TRAC, beam heating, volumetric coronal heating, or radiative cooling stage in the production timestep;
- a fixed lower hydrostatic reservoir and a fixed-pressure upper reservoir;
- separate hydrodynamic and conductive temperature data at the upper boundary;
- deterministic OpenMP execution and validated long-run controls.

The reference runtime table is `data/eos/gamma1_hydrogen_v1.dat`. Loading a Γ_1 table selects the single-fluid release solver, and the `model_column` scenario supplies the production table *unconditionally*; only its path is overridable. `model_column` therefore names exactly one solver and one production model. It refuses `ISO_GAMMA` outright, and the initial-condition builder additionally rejects every historical two-fluid research knob rather than silently accepting it: `ISO_TWO_FLUID`, `ISO_IONIZATION`, `ISO_COOLING`, `ISO_TRAC`, `ISO_CORONA`, `ISO_CHEAT`, `ISO_QFLUX`, `ISO_TBOOST`, and `ISO_NUMERICAL_DIFFUSIVITY_MULT`. The release also rejects RK4, the explicit-only integrator, finite-rate ionization, and the legacy runtime settings `SINGLE_FLUID` and `ENABLE_TE` before evolution begins. A release user therefore needs none of those knobs, and no environment variable can reroute `model_column` into the historical solver. The historical fixed-γ two-fluid column is reached through the separate `model_gentle` scenario, which never loads a Γ_1 table.

2.2 Validated release configuration

The canonical release configuration is:

Item	Reference value
Domain	1600–2153 km, a 553 km truncated upper-chromosphere/lower-TR column
EOS	`data/eos/gamma1_hydrogen_v1.dat`
Coarse-equivalent resolution	`ISO_NS=500`
Static refinement	outer profile, factor 4, fully refined above 2100 km, graded over approximately 2080–2100 km
Actual cell count	661
Fine spacing	approximately 0.276 km at the upper boundary
Hydro top temperature	zero-gradient extrapolation of the live top cell
Conduction top temperature	fixed $T_{\text{face}} = 22000$ K at the physical outer face, a half top-cell width from the top-cell center
Outer pressure	fixed reference back pressure with a one-sided HSE ghost ladder
Outer velocity	live top velocity, capped at $0.1c_s$
Lower boundary	1600 km truncation: isothermal Saha-consistent trapezoidal discrete-HSE reservoir with $v = 0$
Energy physics	physical conduction only; artificial conduction, beam/-coronal heating, radiative cooling, and TRAC are structurally absent from the release operator, not merely disabled
Reconstruction/flux	$(\log \rho, v, \log p)$, MC3/Koren $\beta = 2$, SWMF exact-Riemann Godunov flux at frozen $\gamma = 5/3$
Well balancing	none — reference-free (section 11.4)
Runtime & Lie-split explicit hydrodynamics + implicit conduction; OpenMP launcher defaults to 12 threads	
Validated production CFL	0.50 for this model shape and hardware; solver fallback remains 0.25

Selecting `model_column` supplies these model, grid, and EOS settings as override-preserving scenario defaults; the release has no source stages to configure. The production launcher adds the validated CFL, OpenMP, and low-I/O runtime policy but does not redefine the physical model.

2.3 Capabilities outside the release path

Capability	Status	Disposition
Fixed-γ column	**Historical**	Retained under the separate `model_gentle` scenario; $\gamma = 1.05$ was a useful nearly isothermal surrogate but is not the release thermodynamics. `model_column` rejects `ISO_GAMMA`.
Independent ion and neutral fluids	**Implemented but inactive**	Implemented in the separate two-fluid solver; rejected by the release solver.
Finite-rate collisional/photoionization	**Implemented but inactive**	Implemented for legacy two-fluid studies; replaced by equilibrium Saha ionization in the release path.
Separate electron temperature	**Experimental**	Three-temperature exchange machinery exists in the two-fluid solver; the release is a common-temperature mixture and rejects `ENABLE_TE`.
Optically thick/thin radiation	**Implemented but inactive**	Implemented and tested in the two-fluid solver. Removed from the release timestep; `model_column` rejects `ISO_COOLING`.
TRAC	**Implemented but inactive**	Implemented and tested in the two-fluid solver. Removed from the release conduction operator; `model_column` rejects `ISO_TRAC`.
Artificial (mesh-scaled) conduction	**Historical**	Retained inside the two-fluid Stage-D conduction operator for controlled legacy comparison. Removed from the release operator entirely; `model_column` rejects `ISO_NUMERICAL_DIFFUSIVITY_MULT`.
Imposed heat-flux boundary	**Experimental**	Alternative to the fixed-temperature conductive wall; not the reference boundary.
Volumetric coronal heating	**Experimental**	Implemented for resolved-corona studies in the two-fluid solver; not a release stage.
Nonthermal electron-beam heating	**Experimental**	Implemented for flare experiments in the two-fluid solver; not a release stage.
Loop, flare, gentle, canopy, and PFSS scenarios	**Historical**	Valuable scenario-specific work, not evidence for the current release column.
Multidimensional two-fluid MHD formulation	**Historical**	Preserved as theoretical and architectural background; not evolved by the current one-dimensional release executable.
AWSoM/SWMF coupling	**Planned**	Intended endpoint; no production coupling interface is yet validated.

Part II

Physical Models and Their Evolution

Chapter 3

Active Release Governing Equations

3.1 Closed field-aligned system

Active release path

The release is a one-dimensional hydrodynamic model on a prescribed magnetic field line. The coordinate s follows the field, and magnetic-flux conservation relates the prescribed field strength $B(s)$ to the flux-tube area $A(s)$:

$$A(s)B(s) = \text{const.} \tag{3.1}$$

The magnetic field is geometry, not an evolved component of the state. The release therefore has no induction equation or dynamically evolved magnetic-pressure or Lorentz-force term.

For a time-independent prescribed field and gravitational potential $\Phi(s)$, the continuum equations advanced by the release path are

$$\frac{\partial \rho}{\partial t} + B\frac{\partial}{\partial s}\left(\frac{\rho v}{B}\right) = 0, \tag{3.2}$$

$$\frac{\partial(\rho v)}{\partial t} + B\frac{\partial}{\partial s}\left(\frac{\rho v^2 + p}{B}\right) = pB\frac{\partial B^{-1}}{\partial s} - \rho\frac{\partial \Phi}{\partial s}, \tag{3.3}$$

$$\frac{\partial \mathcal{E}}{\partial t} + B\frac{\partial}{\partial s}\left[\frac{v(\mathcal{E} + p) + q_{\mathrm{c}}}{B}\right] = 0. \tag{3.4}$$

Here

$$\mathcal{E} = e_{\mathrm{int}} + \frac{1}{2}\rho v^2 + \rho\Phi, \tag{3.5}$$

$$q_{\mathrm{c}} = -\kappa_{\mathrm{phys}}(\rho, T)\frac{\partial T}{\partial s}, \qquad \kappa_{\mathrm{phys}} = \kappa_e + \kappa_n. \tag{3.6}$$

The active release has no volumetric heating, radiative cooling, TRAC modification, or artificial conductivity. Its energy input is the physical conductive flux established by the external temperature at the upper physical face.

The equilibrium-mixture closure completes the system. With $n_{\mathrm{H}} = \rho/m_H$, proton and electron fractions $x = n_p/n_{\mathrm{H}} = n_e/n_{\mathrm{H}}$, and hydrogen ionization energy χ_{H},

$$\frac{x^2}{1-x} = \frac{1}{n_{\mathrm{H}}}\left(\frac{2\pi m_e k_B T}{h^2}\right)^{3/2} \exp\left(-\frac{\chi_{\mathrm{H}}}{k_B T}\right), \tag{3.7}$$

$$p = (1+x)n_{\mathrm{H}}k_B T, \tag{3.8}$$

$$e_{\mathrm{int}} = \frac{3}{2}p + x n_{\mathrm{H}}\chi_{\mathrm{H}}. \tag{3.9}$$

The equilibrium acoustic closure of the equation of state is

$$c_{\mathrm{eq}}^2 = \Gamma_1(\rho, T)\frac{p}{\rho}, \tag{3.10}$$

where the production Γ_1 table supplies the equilibrium adiabatic derivative. This is the acoustic response *when the ionization balance re-equilibrates within the wave*. The release numerical flux and the release timestep instead use the frozen-composition response $c_{\mathrm{fr}}^2 = (5/3)\,p/\rho$, on the physical grounds set out in section 11.3.3; both speeds follow from the same closure, and Eq. (3.10) remains the equilibrium derivative of the model. Equations (3.1)–(3.10), together with the prescribed $B(s)$, $\Phi(s)$, initial state, and boundary data, are the self-contained mathematical model. The later thermodynamic, boundary, numerical-flux, and implicit-conduction chapters describe how this closed system is evaluated and discretized.

3.2 Relation between area and magnetic weighting

Using $A \propto B^{-1}$, the continuity operator has the equivalent forms

$$\frac{1}{A}\frac{\partial}{\partial s}(A\rho v) = B\frac{\partial}{\partial s}\left(\frac{\rho v}{B}\right). \tag{3.11}$$

The area form of field-aligned momentum is

$$\frac{\partial(\rho v)}{\partial t} + B\frac{\partial}{\partial s}\left(\frac{\rho v^2}{B}\right) + \frac{\partial p}{\partial s} = -\rho\frac{\partial \Phi}{\partial s}. \tag{3.12}$$

Combining the advective and pressure terms inside one B-weighted flux introduces $pB\partial_s(B^{-1})$. Moving that contribution to the right gives exactly Eq. (3.3). The term is a pressure-area source caused by prescribed tube expansion, not a magnetic force from an evolved field.

The conductive part of Eq. (3.4) is likewise geometry aware:

$$-B\frac{\partial}{\partial s}\left(\frac{q_{\mathrm{c}}}{B}\right) = B\frac{\partial}{\partial s}\left(\frac{\kappa_{\mathrm{phys}}}{B}\frac{\partial T}{\partial s}\right). \tag{3.13}$$

This is the continuum operator represented by the face conductivities and $B_i/B_{i\pm1/2}$ factors in the nonlinear conduction solve.

No separate gravitational source belongs in Eq. (3.4). If $\mathcal{E}_g = e_{\mathrm{int}} + \rho v^2/2$, its equation contains gravitational work $-\rho v\,\partial_s\Phi$. Multiplying Eq. (3.2) by the time-independent Φ gives

$$\frac{\partial(\rho\Phi)}{\partial t} + B\frac{\partial}{\partial s}\left(\frac{\rho v\Phi}{B}\right) = \rho v\frac{\partial \Phi}{\partial s}. \tag{3.14}$$

Adding the two equations cancels gravitational work and produces the conservative total-energy equation written above.

3.3 Constant-field release specialization

Active release path The reported `model_column` calculation sets

$$B_i = B_{i\pm1/2} = 1, \qquad \frac{\partial B^{-1}}{\partial s} = 0. \tag{3.15}$$

The tube area is therefore constant, all B-metric factors reduce to unity, and Eqs. (3.2)–(3.4) become the Cartesian one-dimensional mass, momentum, and total-energy equations. The finite-volume state and the Roe reference eigensystem later abbreviate $\mathcal{E}$ as E; they discretize this specialization of the same closed field-aligned model.

Chapter 4

General Plasma-Neutral Formulation

4.1 Two-fluid model

Historical The project began from a reacting plasma-neutral description in which charged hydrogen and neutral hydrogen have independent mass, momentum, and energy equations. In compact one-dimensional notation,

$$\frac{\partial \rho_s}{\partial t} + \frac{1}{A}\frac{\partial}{\partial s}(A\rho_s u_s) = S_{\rho,s}, \tag{4.1}$$

$$\frac{\partial (\rho_s u_s)}{\partial t} + \frac{1}{A}\frac{\partial}{\partial s}\left[A(\rho_s u_s^2 + p_s)\right] = \rho_s g_s + S_{m,s}, \tag{4.2}$$

$$\frac{\partial E_s}{\partial t} + \frac{1}{A}\frac{\partial}{\partial s}\left[A(E_s + p_s)u_s\right] = \rho_s u_s g_s + S_{E,s}, \tag{4.3}$$

where $s \in \{i, n\}$, $A \propto B^{-1}$ is the flux-tube area, and the source terms include ion-neutral drag, thermal equilibration, ionization, recombination, conduction, radiation, and imposed heating. Related source-consistent formulations are discussed by Meier and Shumlak [2012] and Khomenko et al. [2014].

The full model remains useful when charge-neutral drift, reaction time scales, or independent temperatures are the scientific target. It is not used by the current gamma-table release because that release assumes the strong-coupling, equilibrium-ionization limit.

4.2 Field-aligned reduction and electron folding

Historical The general formulation was reduced from multidimensional MHD to a prescribed field line by evolving parallel flow and treating the field strength as prescribed geometry. Quasi-neutrality gives $n_e = n_i$ for hydrogen. Electron mass is neglected in the total mass and momentum, while electron pressure and thermal conduction remain in the charged-fluid energetics.

In the common-temperature limit, the charged pressure is

$$p_i = (n_p + n_e)k_B T = 2n_i k_B T, \tag{4.4}$$

and the neutral pressure is $p_n = n_n k_B T$. The release mixture pressure is their sum. The field-aligned reduction remains conceptually important for later AWSoM coupling, but the active release uses a straight, constant-area column.

4.3 Ion-neutral drag and thermal equilibration

Implemented but inactive The legacy integrator advances drag through center-of-mass and relative-drift variables. This avoids an explicit collision-time restriction and preserves total momentum. The removed relative kinetic energy is returned to thermal energy. A separate point-implicit temperature stage equilibrates ion, neutral, and optionally electron temperatures.

These stages do not exist in the release solver. Their infinite-coupling limit is instead built into the release state itself: there is one momentum and one internal energy per cell, so a common velocity and a common temperature hold identically at every stage rather than being enforced after the fact.

Chapter 5

Ionization and Radiation Models

5.1 Finite-rate hydrogen network

Implemented but inactive The finite-rate path contains electron-impact ionization, radiative and three-body recombination, a multilevel collisional-radiative channel, and a frozen radiation-field photoionization profile. Representative rates draw on Voronov [1997], Hummer [1994], and the non-equilibrium chromospheric literature [Carlsson and Stein, 2002, Leenaarts et al., 2007].

The reaction stage solves the local backward-Euler composition equation rather than taking an explicit chemistry step. Ionization energy is removed from or returned to the thermal pool according to the reaction channel. This machinery was important in exposing the sensitivity of the upper boundary to energy balance. It is not part of the release solver, whose composition is the Saha equilibrium value of the EOS closure at every instant.

5.2 Radiative closures

Implemented but inactive Two radiative closures are implemented:

1. an optically thick chromospheric loss based on tabulated H I, Ca II, and Mg II escape-probability approximations following Carlsson and Leenaarts [2012]; and
2. an optically thin transition-region/coronal loss $Q_{\text{thin}} = n_e n_H \Lambda(T)$, smoothly activated near 2×10^4 K.

TRAC broadening modifies electron conductivity and optically thin losses below an adaptive cutoff while preserving their product, following Johnston et al. [2019, 2020]. Both closures and TRAC live in the two-fluid research solver; neither is a stage of the release timestep, and `model_column` rejects `ISO_COOLING` and `ISO_TRAC`. Their absence from the release results is therefore a physical limitation of those results, not a missing code capability.

Chapter 6

Equilibrium-Ionization Thermodynamics

6.1 Saha composition

Active release path The current physical core is a classical pure-hydrogen equilibrium mixture. With total hydrogen-nucleus density $n_\mathrm{H} = \rho/m_H$ and ionization fraction $x = n_p/n_\mathrm{H} = n_e/n_\mathrm{H}$,

$$\frac{x^2}{1-x} = \frac{1}{n_\mathrm{H}} \left(\frac{2\pi m_e k_B T}{h^2} \right)^{3/2} \exp\left(-\frac{\chi_\mathrm{H}}{k_B T} \right). \tag{6.1}$$

The implementation evaluates the positive root in logarithmic form to avoid overflow, underflow, and cancellation in the neutral and ionized limits.

All physical thermodynamic quantities use the unclamped solution x_Saha. A small carrier fraction floor is applied only when packing the historical ion and neutral storage rows. Changing that floor changes the trace row but not pressure, temperature, conductivity, radiation, or sound speed.

6.2 Caloric equation of state

Active release path The mixture pressure and internal-energy density are

$$p(\rho, T) = (1 + x_\mathrm{Saha}) n_\mathrm{H} k_B T, \tag{6.2}$$

$$e_\mathrm{int}(\rho, T) = \frac{3}{2} p + x_\mathrm{Saha} n_\mathrm{H} \chi_\mathrm{H}. \tag{6.3}$$

The second term records the energy stored in ionizing hydrogen. In the ionization region, conductive input can therefore change x substantially without producing the temperature rise predicted by a constant-γ gas.

The derivative used by the nonlinear conduction solve is

$$C_V^\mathrm{eq} = \left(\frac{\partial e_\mathrm{int}}{\partial T} \right)_\rho = \frac{3}{2}(1+x) n_\mathrm{H} k_B + \left(\frac{3}{2} n_\mathrm{H} k_B T + n_\mathrm{H} \chi_\mathrm{H} \right) \left(\frac{\partial x}{\partial T} \right)_\rho, \tag{6.4}$$

with

$$\left(\frac{\partial x}{\partial T} \right)_\rho = \frac{x(1-x)}{(2-x)T} \left(\frac{3}{2} + \frac{\chi_\mathrm{H}}{k_B T} \right). \tag{6.5}$$

This effective heat capacity is positive over the production table and becomes large in the partial-ionization region.

6.3 γ_E and Γ_1

Two exponents have different roles and must not be conflated.

Caloric ratio $\gamma_E = 1 + p/e_{\rm int}$ describes how the chosen zero of internal energy affects the ratio of pressure to stored energy. It enters interpretation and diagnostics, not the acoustic eigenvalue.

First adiabatic exponent $\Gamma_1 = (\partial \ln p/\partial \ln \rho)_S$ controls the equilibrium-mixture sound speed,

$$c_{\rm eq}^2 = \Gamma_1 \frac{p}{\rho}. \tag{6.6}$$

Frozen translational exponent $\gamma = 5/3$, the monatomic value, controls the sound speed when the ionization state cannot change on the timescale of the disturbance, $c_{\rm fr}^2 = (5/3)p/\rho$. It follows from the exact split $e_{\rm int} = p/(5/3-1) + e_{\rm ion}$ with $e_{\rm ion}$ held fixed.

Active release path Γ_1 and $\gamma = 5/3$ are the equilibrium and frozen limits of the same closure, not competing approximations to one number. The release face Riemann problem and the release timestep use the frozen limit; Γ_1 remains the tabulated equilibrium derivative and is used for diagnostics, the outer-boundary Mach cap, and the reference Roe flux. Section 11.3.3 gives the physical argument and the literature; section 19.4 records the unresolved consistency question that follows from it.

Active release path The runtime $\Gamma_1(T, n_{\rm H})$ is bilinearly interpolated in $(\log T, \log n_{\rm H})$ from the CRASH-derived table. The caloric closure itself is analytic. This division was deliberate: CRASH supplies the validated acoustic derivative, while pressure, energy, Saha composition, and caloric inversion remain transparent and independently testable.

An independent analytic identity used in validation is

$$\Gamma_1 = \chi_\rho + \frac{p\chi_T^2}{TC_V^{\rm eq}}, \qquad \chi_\rho = \left(\frac{\partial \ln p}{\partial \ln \rho}\right)_T, \quad \chi_T = \left(\frac{\partial \ln p}{\partial \ln T}\right)_\rho. \tag{6.7}$$

6.4 Caloric inversion

Active release path Conserved evolution provides ρ and $e_{\rm int}$, so every decode may require the solution of

$$e_{\rm int}(\rho, T) = e_{\rm int}^{\rm target}. \tag{6.8}$$

The implementation brackets the root on the EOS-table temperature interval, attempts safeguarded Newton updates using $C_V^{\rm eq}$, and falls back to bisection when a Newton step is unsafe or insufficiently reduces the bracket. Out-of-range density or energy is a production error, not a silent clamp. A finite previous temperature is only an accelerator; correctness does not depend on a cache.

Chapter 7

Thermal Conduction and Optional Energy Sources

7.1 Physical conductivity

Active release path The common-temperature conductive flux is

$$q_{\text{phys}} = -(\kappa_e + \kappa_n)\frac{\partial T}{\partial s}. \tag{7.1}$$

The electron coefficient approaches Spitzer behavior in the ionized limit and includes electron-neutral collisions in weakly ionized gas [Spitzer, 1962, Vranjes and Krstic, 2013]. The neutral coefficient dominates much of the lower chromosphere. Physical diagnostic plots use only $\kappa_e + \kappa_n$; they explicitly exclude TRAC and artificial diffusion.

7.2 Numerical conductivity

Historical A stabilizing artificial temperature diffusivity was once folded into the conduction operator:

$$\chi_{\text{num},f} = C_{\text{num}}\,\Delta s_f, \tag{7.2}$$

$$K_{\text{num},f} = \chi_{\text{num},f}\,C^{\text{eq}}_{V,f}, \tag{7.3}$$

with the reference value $C_{\text{num}} = 2000\ \mathrm{m\,s^{-1}}$ before an explicit diagnostic multiplier, and Δs_f the actual center-to-center face spacing. This term is artificial and was never reported as physical conductivity.

It has been removed from the release conduction operator entirely: the release face conductivity is $\kappa_e + \kappa_n$ and there is no $K_{\text{num},f}$ term to disable. The release is therefore physical-conduction-only structurally, not because a multiplier happens to be zero, and `model_column` rejects `ISO_NUMERICAL_DIFFUSIVITY_MULT`. Equation (7.3) remains implemented inside the two-fluid Stage-D operator so that the historical convergence studies recorded in this document can be reproduced.

7.3 Volumetric heating and cooling stages

Implemented but inactive Beam heating, volumetric coronal heating, and radiative cooling are implemented as post-conduction source stages of the *two-fluid* solver. Each modifies only the thermal pool of a cell. They are not stages of the release timestep, which consists of the explicit hydrodynamic update and the implicit physical conduction solve and nothing else; `model_column` rejects `ISO_COOLING`, `ISO_CHEAT`, and the flare beam configuration.

Experimental An imposed outer heat flux can replace the fixed-temperature wall, and a resolved-corona option can combine TRAC, radiation, and volumetric heating in the two-fluid solver. These configurations are useful for future physical studies but are not the current release baseline.

Part III

Current `model_column` Implementation

Chapter 8

Geometry, Domain, and Initial State

8.1 Geometry and gravity

Active release path The release column is vertical, straight, and constant area: $B_i = B_{i\pm1/2} = 1$ and $\partial_s(B^{-1}) = 0$. The code retains general flux-tube metric factors, but they reduce to unity in the active configuration. Gravitational potential is stored at faces and centered for cell decoding. The acceleration magnitude is 273.95 $\mathrm{m\,s^{-2}}$.

8.2 Model C7 temperature data

Active release path Model C7 provides the temperature profile and the base total hydrogen density. The complete repository table spans the photosphere through the corona, but the release initializer samples only the requested domain.

The historical linear temperature interpolation was continuous but had a piecewise-constant derivative, which seeded step-like conductive-flux structure at every sparse C7 knot. The gamma-table initializer now uses a Fritsch–Carlson/Fritsch–Butland shape-preserving cubic Hermite interpolant. It passes exactly through the C7 knots, remains monotone on monotone intervals, and is C^1 across internal knots. The fixed-gamma scenarios retain their historical interpolation for regression stability.

8.3 Saha-consistent hydrostatic reintegration

Active release path The release does not use the tabulated C7 electron and neutral densities directly away from the base. It retains $T(h)$, anchors the base with the C7 total hydrogen density, and reintegrates hydrostatic pressure according to the active Saha EOS.

Between cell centers,

$$\ln p_i = \ln p_{i-1} - \frac{\Delta h_i}{2}\left(H_{i-1}^{-1} + H_i^{-1}\right), \qquad H^{-1} = \frac{m_H g}{(1+x)k_B T}. \tag{8.1}$$

Because x depends on the density implied by p, each cell solves a pressure-density fixed point. Non-convergence after 60 iterations is a hard error with cell, height, temperature, and pressure context.

The resulting cell state is packed directly on the equilibrium manifold. No frozen copy is retained: the release keeps no reference equilibrium (section 11.4), and `grid.eq_state` is populated only on the non-release two-fluid leg of this scenario.

Chapter 9

Boundary Conditions

9.1 Two ghost cells

Active release path Every boundary provides two packed ghost states because the MUSCL reconstruction accesses two exterior layers. Each ghost is packed with the gravitational potential used at that boundary face. The ghosts are not arbitrary extrapolation buffers: they are part of the discrete boundary closure and must satisfy the same EOS as interior face states.

9.2 Release boundary summary

Active release path The release has two distinct boundary mechanisms:

> **Hydrodynamic ghost states** close the MUSCL reconstruction stencil and contain pressure, hydrodynamic temperature, EOS-derived composition, and packed energy. **Conductive physical-face data** close the nonlinear implicit conduction operator. The fixed 22000 K upper conductive datum is not a hydro ghost temperature, and the second upper hydro ghost has no conductive role.

Quantity	Lower boundary	Upper boundary	BC type / independent?	Numerical implementation
u	Fixed reservoir, $u = 0$	Live top-cell velocity with $\lvert u_g\rvert \leq 0.1c_{\mathrm{eq}}$	Lower independent; upper extrapolated and capped	Stored in both hydro ghosts used by the MUSCL stencil
p	Two-rung discrete-HSE continuation	First ghost fixed to p_{ref}; second from one-sided HSE	Upper p_{ref} is the single external hyperbolic datum	Ghost pressure is used with T_{hydro} to close the EOS
T_{hydro}	Fixed reservoir temperature	Zero-gradient from the live top cell	Lower independent; upper extrapolated	Used only to construct hydro ghost density, composition, energy, and Riemann states
ρ	EOS recovery from HSE pressure and fixed T	EOS recovery from ghost pressure and T_{hydro}	Derived at both boundaries	$\rho = \rho(p, T)$ from the equilibrium hydrogen closure

Quantity	Lower boundary	Upper boundary	BC type / independent?	Numerical implementation
x	Saha equilibrium at the ghost (ρ, T)	Saha equilibrium at the ghost $(\rho, T_{\mathrm{hydro}})$	Derived at both boundaries	Recomputed by the equilibrium face-state packer
Energy	Packed from the reservoir thermodynamic state	Packed from each EOS-derived hydro ghost state	Derived; not separately prescribed	Preserves the mixture energy convention, including ionization and gravitational energy
Conductive T	Fixed lower reservoir temperature	$T_{\mathrm{face}} = 22000$ K at the physical outer face	Independent Dirichlet data	Lower state enters the left conduction face; upper state is separate from both hydro ghosts
Conductive heat flux	Determined by the live first cell and reservoir state	Determined by the live top cell and physical-face state	Derived, not independently imposed	$q_{\mathrm{top}} = \kappa_{\mathrm{face}}(T_{\mathrm{face}} - T_{\mathrm{top}})/(\Delta s_{\mathrm{top}}/2)$

9.3 Lower truncation of the modeled domain

Active release path **The lower boundary at 1600 km is not the photosphere.** It is a deliberate truncation inside the chromosphere, placed at the height above which this solver's Saha/LTE equilibrium closure is intended to be used. The atmosphere below it is real and is simply outside the model. Three things must therefore be kept apart:

Physical boundary information at 1600 km. On the timescales resolved here, the chromosphere below the truncation acts as a quasi-static stratified reservoir. That is the entire physical content of the condition: it fixes a reservoir temperature T_0, a hydrostatic pressure stratification, and $v = 0$.

Numerical ghost closure below it. MUSCL reconstruction needs two exterior layers, so two ghost states are packed one and two base-cell widths below cell 0. They are a stencil closure. They are *not* an assertion that LTE, or this model, describes the atmosphere at those heights, and no physical meaning should be read into their densities or ionization fractions.

Dependence on a global equilibrium. There is none. The reservoir reference is captured once from cell 0 of the initial condition and is a local base-cell quantity; the release does not hold a global hydrostatic profile (section 11.4).

Each ghost pressure is a trapezoidal, EOS-closed hydrostatic step taken downward from the rung above,

$$p_{k+1} = p_k + \Delta s_0\, \tfrac{1}{2}(\rho_k + \rho_{k+1})\, g, \qquad \rho_{k+1} = \rho_{\mathrm{EOS}}(p_{k+1}, T_0)\,, \tag{9.1}$$

iterated to a fixed point, with Δs_0 the actual base-cell width (on the refined mesh this is the fine spacing, not the coarse one). Ghost densities come from the equilibrium EOS rather than being copied flat, which keeps the pressure and logarithmic-density slopes mutually consistent and removes the Rusanov mass leak of the older flat-density boundary.

Retired Equation (9.1) replaces the earlier first-order ladder $p_{g,1} = p_0 + \rho_0 g \Delta s_0$, $p_{g,2} = p_0 + 2\rho_0 g \Delta s_0$, which used the cell-0 density for both rungs. That form carried an $\mathcal{O}((\Delta s_0/H)^2)$ pressure error and hence an $\mathcal{O}(\Delta s_0/H)$ error in the hydrostatic force balance of the lowest interior cells, and it was inconsistent with the trapezoidal ladder the outer face had already been using. With the predictor source of section 11.2.1 in place but the old ladder still present, the reference-free 20 s residual velocity peaked at 1607.7 km — in the first few interior cells. Equation (9.1) removed that layer: the base-band max $|v|$ fell from 3.2×10^{-2} to 1.4×10^{-2} m s^{-1}, the global maximum moved to the upper transition region, and the $t = 0$ face mass-flux defect fell by a further factor of 11.

Historical Because 1600 km is an artificial cut rather than a wall, a fixed $v = 0$ reservoir is in principle an acoustic mirror, and a gravity-aware characteristic alternative was built and measured. It imposes only the incoming (upward, $u + c$) acoustic invariant $\delta u + \delta p/Z = 0$ about the base reference state with impedance $Z = \rho c$, taking the outgoing invariant from the live base cell, which gives ghost deviations $v_g = -(p_0 - p_{0,\mathrm{ref}})/Z$ and $p_g = p_{g,\mathrm{ref}} - Zv_0$. It degenerates exactly to the fixed reservoir at equilibrium, so it costs nothing in hydrostatic balance. It also gained nothing: over 20 s hydrostatic, 100 s and 1000 s conduction-driven runs it left the $t = 0$ defect and the residual-velocity profile unchanged, altered the evaporation-window velocity by $\leq 0.4\%$, and left the detrended base-region oscillation content unchanged (3 sign changes in 150 s either way; peak-to-peak 0.31 vs 0.31 $\mathrm{m\,s^{-1}}$). The disturbance the transition region sends down arrives as a slow quasi-static pressure adjustment, not as acoustic ringing, so this face is not an important reflector for this model. The characteristic condition was therefore not promoted, and its code was removed rather than left as a default-off switch. **Limitation:** the corollary is that mass cannot cross 1600 km, so the column cannot be replenished from below during evaporation — the characteristic variant recovered about 17% of the 1000 s mass loss this way. That is a modeling choice of the truncation, not a numerical defect, and it should be revisited if base mass supply becomes a quantity of interest.

The lower conduction condition is separate. The default is a fixed-temperature reservoir; `ISO_INNER_T_NEUMANN=1` instead imposes zero conductive flux. Either choice leaves the hydrodynamic closure at $v = 0$.

9.4 Upper hydrodynamic reservoir

Active release path The upper boundary represents an external reservoir through one imposed hydrodynamic datum: a fixed reference back pressure. This is appropriate for subsonic outflow, which has one incoming characteristic.

At initialization, or after an optional relaxation stage, the code captures a reference pressure one one-sided hydrostatic step outside the top cell. That pressure is then held fixed. The second ghost is obtained from another one-sided HSE rung,

$$p_{k+1} = p_k - \frac{\Delta s_N g}{2}(\rho_k + \rho_{k+1}), \tag{9.2}$$

with the density closed by the equilibrium EOS at the hydrodynamic ghost temperature. The local fixed-point iteration is repeated twice, sufficient for the small hydrostatic correction.

The ghost velocity copies the live top-cell velocity but is clipped to

$$|v_g| \leq M_{\mathrm{cap}} c_s, \qquad M_{\mathrm{cap}} = 0.1 \tag{9.3}$$

in the reference configuration. During an optional relaxation phase it is fixed to zero.

9.5 Hydrodynamic and conductive temperatures

Active release path The upper boundary deliberately carries two temperatures:

Hydrodynamic ghost temperature zero-gradient extrapolation from the live top cell when `ISO_HYDRO_T_DECOUPLE=1`; this temperature determines ghost density, internal energy, and the Riemann state.

Conductive physical-face temperature a fixed external value, 22000 K in the release calculation; this appears only in the implicit conduction stencil.

This separation avoids imposing both pressure and temperature on the hydrodynamic subsystem while retaining a controlled thermal reservoir. The conductive Dirichlet datum is located at the physical outer face. For the top cell,

$$d_{\mathrm{boundary}} = \frac{1}{2}\Delta s_{\mathrm{top}}, \qquad q_{\mathrm{top}} = \kappa_{\mathrm{face}} \frac{T_{\mathrm{face}} - T_{\mathrm{top}}}{d_{\mathrm{boundary}}}, \qquad T_{\mathrm{face}} = 22000\ \mathrm{K}. \tag{9.4}$$

The face conductivity is evaluated from the live top-cell state and an external Saha-equilibrium face state constructed from the fixed back pressure and T_{face}. The hydro solver still reconstructs through two ghost cells, but neither the first- nor second-ghost center defines the release conductive wall.

9.6 Boundary-condition development history

The upper boundary was revised by numerical question rather than by date:

Question	Controlled change	Result	Disposition
Could centered pressure extrapolation preserve HSE?	Anchor the first ghost from the second interior cell across the top cell.	In the non-hydrostatic upper TR it inverted the boundary pressure gradient and produced a standing downward force and last-cell reversal.	Retired
Could a live top-cell pressure anchor be transparent?	Use the current top pressure and one-sided HSE outside it.	Removed back pressure; the subsonic column drained, accelerated to about 1.9 km s^{-1}, and failed near 12 s.	Retired
What supplies the incoming characteristic?	Capture and hold a fixed external reference pressure.	Stable reservoir pressure without gradient inversion or uncontrolled drainage.	Active release path
Was hydro temperature over-specified?	Keep the 22 kK conduction wall but set hydro ghost temperature from the top cell.	Removed the last-cell velocity reversal; the broader interior ripple remained.	Active release path
Was the remaining ripple a flux discontinuity?	Record central, diffusive and total face fluxes against the $t = 0$ baseline.	The effective conserved face flux was smooth; cell-centered ρv and Rusanov components were anti-correlated.	Diagnostic conclusion

Does the ripple converge?	Uniform and local resolution scans.	Ripple/mismatch decreases approximately first order and is efficiently reduced by upper-TR refinement.	Validated numerical interpretation
Was the old ghost-center conductive input converged?	N=1000/2000/4000 with local artificial conductivity.	Mass flux changed only 2% on the last pair, but total conductive input still changed 19%.	Historical Superseded by physical-face, physical-only conduction

Part IV

Numerical Method

Chapter 10

Conserved and Primitive States

10.1 Authoritative release state

Active release path The release solver stores and advances exactly three conserved rows per cell,

$$U = (\rho,\ m,\ E)^{\mathsf{T}}, \qquad m = \rho v, \qquad E = e_{\text{int}}(\rho, T) + \frac{m^2}{2\rho} + \rho\Phi, \tag{10.1}$$

in the row order `mix::RHO`, `mix::MOM`, `mix::ENERGY` of `chromosphere.hpp`. This is the discrete form of Eqs. (3.2)–(3.5): nothing else is stored, and every stage of the timestep — reconstruction, numerical flux, geometric source, boundary ghosts, implicit conduction, and the timestep control — reads and writes these three rows directly.

Everything else is a derived thermodynamic quantity of (ρ, e_{int}) produced by the equilibrium closure:

$$\begin{aligned}
x &= x_{\text{Saha}}(n_{\text{H}}, T), & n_{\text{H}} &= \rho/m_H, && (10.2)\\
n_e &= x n_{\text{H}}, & n_{\text{HI}} &= (1-x)n_{\text{H}}, && (10.3)\\
\rho_i &= x\rho, & \rho_n &= (1-x)\rho, && (10.4)\\
p &= (1+x)n_{\text{H}}k_B T, & p_e &= x n_{\text{H}} k_B T. && (10.5)
\end{aligned}$$

They are single-valued functions of the conserved state, they are recomputed wherever needed, and none of them is an independent degree of freedom. In particular there is no ion/neutral mass split, no per-carrier momentum, and no electron energy row in the release state.

Historical Earlier releases stored the same physics in the seven-row carrier vector $(\rho_i, \rho_n, \rho_i v_i, \rho_n v_n, E_i, E_n, E_e)$ inherited from the two-fluid solver, with the ionization fraction clipped to a trace floor $f_{\min} = 10^{-8}$ so that both mass rows stayed representable. That layout carried four redundant degrees of freedom for a three-degree-of-freedom physical system, which is why it required a per-step conservative repacking (below). The seven-row state now belongs solely to the non-release two-fluid solver of Chapter 4.

10.2 Retired conservative equilibrium projection

Retired Because the seven carrier rows were fluxed independently, an explicit hydrodynamic step left them slightly off the equilibrium manifold: the per-carrier momenta drifted apart and the mass split no longer matched x_{Saha}. A projection stage therefore ran after every explicit update. It formed

$$\rho = \rho_i + \rho_n, \qquad m = m_i + m_n, \qquad v_{\text{cm}} = m/\rho, \tag{10.6}$$

$$e_{\text{int}}^{\text{proj}} = E_i + E_n - \frac{m^2}{2\rho} - \rho\Phi, \tag{10.7}$$

inverted $e_{\text{int}}(\rho, T)$, evaluated Saha equilibrium, and rebuilt the rows, thermalizing the removed relative-drift kinetic energy $\frac{1}{2}(\rho_i\rho_n/\rho)(v_i - v_n)^2$ exactly once.

The three totals (ρ, m, E_i+E_n) were invariant under that map — the projection was the identity on the physical state and acted only on the redundant rows. With the native three-variable state of Eq. (10.1) there is no redundancy to repair, so the stage has been removed entirely: it is not an active release stage, and the spurious per-step drift heating it used to thermalize no longer exists. The removal also eliminates one full caloric inversion per cell per step.

Chapter 11

Finite-Volume Hydrodynamics

11.1 Nonuniform control volumes

Active release path Cell widths, center separations, and face metrics are cached explicitly. For a prescribed area A, a conserved state obeys

$$\frac{\mathrm{d}U_i}{\mathrm{d}t} = -\frac{1}{A_i \Delta s_i}\left(A_{i+1/2}F_{i+1/2} - A_{i-1/2}F_{i-1/2}\right) + S_i. \tag{11.1}$$

The active straight column has constant area, but the metric-aware implementation is retained. Uniform meshes follow guarded legacy algebra to preserve byte-level regression behavior.

11.2 Equilibrium-manifold MUSCL reconstruction

Active release path The release branch decodes each cell through the equilibrium closure and reconstructs the primitive triple

$$W = (\log\rho,\ v,\ \log p). \tag{11.2}$$

Logarithmic density and pressure enforce positivity without independent species clipping, and the face temperature is recovered by inverting the same authoritative Saha closure, $T(\rho, p)$. Each face state is then rebuilt algebraically on the Saha equilibrium manifold.

Historical The release previously reconstructed $W = (\log\rho,\ v,\ \log T)$ and obtained the face pressure afterwards from the nonlinear map $p(\rho, T)$. Two independently limited variables then set one mechanical quantity, so a mechanically smooth (constant-p) state was not reproduced across the partial-ionization transition. The resulting artificial face-pressure mismatch was the dominant source of the late-time coarse-grid velocity ripple; switching to $\log p$ reduced the mismatch by two to three orders of magnitude under both numerical fluxes, so it is a reconstruction improvement rather than a solver-specific patch. The temperature form remains selectable as a reference configuration (`ISO_RECONSTRUCTION=lnrho-v-lnt`) and is not a release setting.

On nonuniform grids, slope ratios are formed from center-to-center gradients, and each extrapolation uses the owning cell's half width. The active limiter is the asymmetric MC3/Koren form with $\beta = 2$. A diagnostic override can select minmod or first-order reconstruction, but that override is not a release setting.

11.2.1 The MUSCL–Hancock step

Active release path The hydrodynamic time discretization is the MUSCL–Hancock method [van Leer, 1979, Toro, 2009]; Berthon [2006] gives the form closest to what is implemented here, including the use of primitive rather than conservative gradients in the reconstruction. Naming matters because "MUSCL" alone describes only the spatial reconstruction. The three pieces are:

MUSCL reconstruction. Limited extrapolation of $W = (\log\rho,\ v,\ \log p)$ from each cell centre to its two faces, as described above.

Hancock predictor. Each cell is advanced to the half-time level using *only its own* two extrapolated fluxes, plus the momentum source:

$$U_i^p = U_i^n - \frac{\Delta t}{\Delta s_i}\Big[F\big(W^L_{i+1/2}\big) - F\big(W^R_{i-1/2}\big)\Big] + \Delta t\, S(U_i^n)\,, \tag{11.3}$$

with $S = \big(0,\ p\,B\,\partial_s B^{-1} - \rho\,\partial_s\Phi,\ 0\big)^\top$. This stage needs no neighbour data and therefore no Riemann solve; only two of the four face reconstructions are built for it. The half state is decoded through the *caloric* EOS only — it never needs the acoustic Γ_1 — and legacy `cons2prim` is not used.

Corrector. The four face states are rebuilt from the time average $\frac{1}{2}(W^n + W^p)$, and the release Godunov flux of section 11.3.2 is evaluated on those time-centred states.

The source term in Eq. (11.3) is not optional. The bracketed flux difference is cell i's own pressure gradient, which in a hydrostatic column is balanced by gravity; omitting S therefore leaves the half state of a column at rest with a spurious velocity of order $\Delta t\,|g|$, and the corrector reconstructs that velocity into every face. Measured on the release mesh with no equilibrium reference, restoring S cut the $t = 0$ discrete hydrostatic face mass-flux defect from 7.5×10^{-10} to $1.4\times10^{-11}\,\mathrm{kg\,m^{-2}\,s^{-1}}$, a factor of 52, and the 20 s conduction-off max $|v|$ from 0.71 to $0.032\,\mathrm{m\,s^{-1}}$. Because $\rho\Phi$ is carried inside $\mathcal{E}$, gravity contributes no energy-row source in either stage, and for the constant-area release geometry $B \equiv 1$ so the flux-tube term vanishes identically; it is retained because the operator is written for general $B(s)$.

Historical A separate experiment restructured the momentum equation in the manner of SWMF's `ModTransitionRegion`: remove the pressure from the momentum flux entirely and evaluate its gradient explicitly alongside gravity, using the acoustic star pressure of the reconstructed face states. On this constant-area, ultra-low-Mach column it reproduced the predictor-source result and nothing more, and it was rejected on that evidence alone. For $B \equiv 1$ the shared face pressure telescopes and the momentum update stays conservative, so conservation is not a reason to reject it here; the form is, however, not generally equivalent to the conservative area-weighted operator on a variable-area flux tube. The useful part of that experiment was the isolation of Eq. (11.3)'s missing source term.

11.3 Numerical flux

Active release path Every face solver in this chapter acts on the authoritative equilibrium-mixture system

$$U = (\rho, m, E)^\mathsf{T}, \qquad m = \rho v, \qquad F(U) = \begin{pmatrix} m \\ m^2/\rho + p \\ (m/\rho)(E+p) \end{pmatrix}, \tag{11.4}$$

where E is the stored energy row of Eq. (10.1), including mixture internal, kinetic, and gravitational energy. Both reconstructed states at a face use the same face gravitational potential, so gravity does not enter the state jump seen by the characteristic dissipation.

11.3.1 Reference flux: the mixture Roe characteristic linearization

Implemented but inactive *Not the release flux.* The mixture Roe characteristic flux was the release numerical flux until the Godunov cutover of section 11.3.2, and is retained as the controlled comparison solver, selected by `ISO_RIEMANN=roe-local`. It is documented here in full because it is the reference against which the release flux is quantified, and because selecting it also reverts the timestep to the equilibrium signal speed (section 12.2), so the comparison mode remains internally consistent.

Its defining property, for the present purpose, is that it linearizes the *equilibrium* system: its acoustic eigenvalues carry Γ_1, so it is the discrete realization of the instantaneous-Saha-equilibrium acoustic closure discussed in section 11.3.3.

For a local equilibrium state define

$$b = \left(\frac{\partial p}{\partial e_{\text{int}}}\right)_\rho, \qquad c^2 = \Gamma_1 \frac{p}{\rho}, \qquad H = \frac{E+p}{\rho}. \tag{11.5}$$

The implemented right eigensystem is

$$\lambda_- = v - c, \qquad r_- = \begin{pmatrix} 1 \\ v-c \\ H-vc \end{pmatrix}, \tag{11.6}$$

$$\lambda_0 = v, \qquad r_0 = \begin{pmatrix} 1 \\ v \\ H-c^2/b \end{pmatrix}, \tag{11.7}$$

$$\lambda_+ = v + c, \qquad r_+ = \begin{pmatrix} 1 \\ v+c \\ H+vc \end{pmatrix}. \tag{11.8}$$

The contact-wave energy component $H - c^2/b$ carries the non-ideal-EOS correction; it reduces to the usual ideal-gas form when the corresponding ideal-gas derivative is inserted.

At each face, the code evaluates $v_q = m_q/\rho_q$, $p_q = F_{m,q} - \rho_q v_q^2$, $H_q = (E_q + p_q)/\rho_q$, b_q, and $c_q^2 = \Gamma_{1,q} p_q/\rho_q$ for $q = L, R$. With $w_q = \sqrt{\rho_q}$, it then forms

$$\widetilde{v} = \frac{w_L v_L + w_R v_R}{w_L + w_R}, \qquad \widetilde{H} = \frac{w_L H_L + w_R H_R}{w_L + w_R}, \tag{11.9}$$

$$\widetilde{b} = \frac{w_L b_L + w_R b_R}{w_L + w_R}, \qquad \widetilde{c}^2 = \frac{w_L c_L^2 + w_R c_R^2}{w_L + w_R}. \tag{11.10}$$

All quantities below are evaluated at this averaged state, with tildes suppressed for readability.

For

$$\Delta U = (\Delta\rho, \Delta m, \Delta E)^{\mathsf{T}}, \tag{11.11}$$

the implemented decomposition first forms

$$\theta = \frac{b}{c^2}\left[\Delta E - v\Delta m + (v^2 - H)\Delta\rho\right], \qquad \eta = \frac{\Delta m - v\Delta\rho}{c}, \tag{11.12}$$

and then

$$\alpha_- = \frac{1}{2}(\Delta\rho + \theta - \eta), \qquad \alpha_0 = -\theta, \qquad \alpha_+ = \frac{1}{2}(\Delta\rho + \theta + \eta). \tag{11.13}$$

Thus the characteristic dissipation and numerical flux are

$$D = \sum_{k\in\{-,0,+\}} |\lambda_k|\alpha_k r_k, \qquad F^{\mathrm{Roe}} = \frac{1}{2}\left(F_L + F_R - D\right). \tag{11.14}$$

The three components of D are applied directly to the three conserved rows. There is no carrier-row redistribution of the characteristic dissipation and no subsequent repacking: the flux written by the solver is exactly the flux in the equation above.

This is a local mixture Roe characteristic linearization. It is not claimed to define an exact nonlinear general-EOS Roe matrix satisfying

$$F_R - F_L = \widetilde{A}(U_R - U_L). \tag{11.15}$$

If either state has invalid density, pressure, enthalpy, sound speed, or b, or if the constructed dissipation is non-finite, only that face falls back to Rusanov.

Implemented but inactive The Rusanov / local Lax–Friedrichs flux,

$$F_{i+1/2} = \frac{1}{2}\left[F(U_L) + F(U_R)\right] - \frac{a_{i+1/2}}{2}(U_R - U_L), \tag{11.16}$$

with

$$a_{i+1/2} = \max\left(|v_L| + c_{s,L}, |v_R| + c_{s,R}\right), \qquad c_s^2 = \Gamma_1 p/\rho, \tag{11.17}$$

remains available as a reference solver (`ISO_RIEMANN=rusanov`) and as the per-face fallback of *both* other solvers. It was the release flux two numerical-method generations ago. Its dissipation is acoustic in scale, $\sim \frac{1}{2}c_s\Delta U$, while the evaporation flow satisfies $|v| \ll c_s$; the resulting numerical transport left a persistent cell-centered velocity and momentum ripple in the upper chromosphere and transition region that both the Roe and the Godunov flux remove. Note that $a_{i+1/2}$ retains the *equilibrium* speed $c_s^2 = \Gamma_1 p/\rho$ even when it serves as the fallback of the frozen-composition release flux: on a fallback face the exact solve has been abandoned and the physical wave speeds of the equilibrium system are the correct Rusanov coefficient. The release timestep is the smaller of the two bounds in any case, so the fallback is never under-resolved in time.

11.3.2 Release flux: the SWMF-style exact-Riemann Godunov flux at frozen composition

Active release path This is the release numerical flux. It follows the construction SWMF uses in `util/CRASH/src/test_godunov.f90` with the exact ideal-gas Riemann solver of `share/Library/src/ModExactRS` (Toro's exact two-rarefaction/two-shock pressure iteration, Toro, 2009), ported line for line to `src/single_fluid/exact_rs.hpp`. It requires no environment variable; `ISO_RIEMANN=roe-local` and `ISO_RIEMANN=rusanov` select the two reference solvers instead. The three are mutually exclusive, so the face Riemann solver is the sole variable the setting changes — reconstruction, the MUSCL–Hancock predictor of section 11.2.1, the momentum source, the boundary closures, the Saha closure and the conduction split are identical under all three — with the single deliberate exception of the timestep, which follows the flux in force (section 12.2).

The method solves an *ideal-gas* Riemann problem at a fixed $\gamma = 5/3$ and carries every non-ideal part of the energy in a passive specific offset attached to the material. For a face state (ρ, u, p, E),

$$E_0 = \frac{E - \frac{1}{2}\rho u^2 - p/(\gamma-1)}{\rho} = \frac{e_{\mathrm{int}} - p/(\gamma-1)}{\rho} + \phi_g, \tag{11.18}$$

so that E_0 absorbs both the Saha ionization energy inside e_{int} and the gravitational potential this model carries in E. Because both sides of a face share ϕ_g, only the equation-of-state content is actually upwinded. With $(\rho_\star, u_\star, p_\star)$ the exact self-similar solution sampled along $x/t = 0$, and E_0 taken from whichever side the contact declares upwind,

$$E_\star = \frac{p_\star}{\gamma-1} + \tfrac{1}{2}\rho_\star u_\star^2 + \rho_\star E_0, \qquad F^{\mathrm{God}}_{i+1/2} = \big(\rho_\star u_\star,\ \rho_\star u_\star^2 + p_\star,\ (E_\star + p_\star)u_\star\big). \tag{11.19}$$

When the sampled state coincides with a side state this reproduces that side's total energy exactly, and a constant offset is advected without altering (ρ, u, p) at all. Any face whose exact solve fails — inadmissible input, the two-rarefaction vacuum guard, a non-positive star pressure, or an exhausted iteration budget — or whose sample is inadmissible falls back to the same face-local Rusanov flux the Roe path uses, and is counted and reported at the end of the run rather than absorbed silently.

The solver reproduces the five standard Riemann tests of Toro [2009] to the six significant digits their star pressures are tabulated to, and a first-order Godunov solve of Sod's problem with equation (11.19) converges at the expected L^1 rate while leaving a nonzero E_0 inert to round-off. On the conduction-off $N = 500$/R4 release mesh (661 cells, CFL 0.50, 3620 steps) the initialized-atmosphere face mass-flux defect is 1.08×10^{-12} against 1.33×10^{-12} for Roe, and the 20 s velocity residual is $3.19 \times 10^{-2}\,\mathrm{m\,s^{-1}}$ against 3.02×10^{-2} — both at the same stored-precision floor, so the measurement does not separate them. On the separate 100 s release smoke configuration (17781 steps, identical for both solvers) every conduction-driven observable agrees to a few parts in 10^4: mean upflow 14.96 against $14.96\,\mathrm{m\,s^{-1}}$, effective face mass flux 1.1168×10^{-9} against $1.1165 \times 10^{-9}\,\mathrm{kg\,m^{-2}\,s^{-1}}$, with identical T_{top}, p_{top}, boundary-layer thickness and conductive flux, and no negative-velocity cells in either. The exact solve succeeded at all 2.35×10^7 face evaluations with no fallback and with the star-pressure Newton iteration converging in a single step at every face, at a cost of +12.9 % in the explicit right-hand side and +4 % in measured whole-step work. The dissipative part of the Godunov mass flux runs 8–12 % below Roe's during the evaporation phase; no causal attribution is claimed for that, because the measurement does not isolate the contact field and a competing effect — the frozen flux's faster waves — pushes the opposite way.

Active release path Every release run reports what the exact solver did. The driver prints a `godunov.*` tally (faces attempted, faces solved exactly, fallbacks broken down by cause, and Newton-iteration statistics), warns on `stderr` if any face fell back, and `scripts/release_validation.sh` fails the release smoke on a nonzero fallback count. A silent degradation to Rusanov is therefore not possible.

11.3.3 Thermodynamic interpretation: frozen composition versus instantaneous Saha equilibrium

Active release path The fixed $\gamma = 5/3$ of Eqs. (11.18)–(11.19) is the substantive modelling content of this flux, and it is a statement about the thermodynamics of the wave, not an ideal-gas approximation to the equation of state. This subsection states the distinction, because it is easy to state it wrongly.

The release caloric closure, Eq. (3.9), splits exactly into a translational term and an ionization reservoir,

$$e_{\mathrm{int}} = \frac{p}{5/3 - 1} + e_{\mathrm{ion}}, \qquad e_{\mathrm{ion}} = x n_{\mathrm{H}} \chi_{\mathrm{H}}, \tag{11.20}$$

so a monatomic translational gas plus a store of ionization energy is not an approximation of this equation of state — it is an identity satisfied by it. The two closures differ only in what happens to e_{ion} while a compressive disturbance passes.

Instantaneous Saha equilibrium. If the ionization balance re-equilibrates within the wave, the isentropic response is the equilibrium first adiabatic exponent Γ_1 of equation (3.10), and the acoustic speed is $c_{\mathrm{eq}}^2 = \Gamma_1 p/\rho$. Compressing the gas then partly ionizes it, the ionization reservoir absorbs part of the compressional work, and Γ_1 falls well below the monatomic value: in the release column its median is 1.090 and it lies below 1.5 across 94.9 % of cells.

Frozen composition. If the ionization state cannot change on the timescale of the wave, e_{ion} is inert, only the translational term of equation (11.20) responds, and the acoustic speed is the monatomic frozen value $c_{\mathrm{fr}}^2 = (5/3)\, p/\rho$.

Neither exponent is universally correct; they are the two limits of the same closure, and the choice between them is set by the ratio of the ionization/recombination relaxation time to the dynamical time of the disturbance. For chromospheric hydrogen that ratio is decisively in favour of the frozen limit. Carlsson and Stein [2002] compute the relaxation timescale directly and find it dominated by the slow collisional leakage from the ground state to $n = 2$: about 1 s in the photosphere, rising to $\sim 10^5$ s in the mid-chromosphere and falling to $\sim 10^2$ s at the transition-region base, and still 10–10^3 s inside shocks, so that "the timescale is still too long for the ionization to reach its equilibrium value in the shock". Leenaarts et al. [2007] reach the same conclusion in two-dimensional radiation-MHD and state the energetic consequence directly — in non-equilibrium, hydrogen ionization is "a less effective internal energy buffer", giving shock temperatures of 10 000–13 000 K against $\leq$ 8000 K under LTE. Leenaarts [2020] puts the chromospheric hydrodynamic timescale at about one minute and summarizes the limit as: with a long relaxation time "an increase in internal energy leads directly to a temperature increase", and "the gas in the chromosphere behaves somewhat like an ideal gas".

The release therefore adopts the frozen-composition interpretation for the local Riemann interaction: the composition is held fixed while the waves cross the face, the ionization energy is carried inertly in E_0, the translational gas is monatomic, and the exact Godunov solver evolves precisely that frozen-composition local problem. Γ_1 is not discarded. It remains the equilibrium derivative of the closure, it is the quantity reported in the diagnostic sidecar, it sets the outer-boundary Mach cap, and it is what the reference Roe flux of section 11.3.1 linearizes.

Two things this argument does *not* establish should be stated plainly. It does not show that the release result is more accurate: no observable measured on this model separates the two closures, and the numerical case for the change is neutral (section 11.3.2). And it does not resolve the consistency of the frozen Riemann solve with the equilibrium decode applied around it — see section 19.4.

11.3.4 Remaining limitations of the release flux

Active release path Three limitations survive the promotion, and one was retired by it.

1. **The two closures have not been separated by measurement on this model.** At the Mach numbers of the release column, every physical observable agrees between the frozen and equilibrium fluxes to a few parts in 10^4. The release choice rests on the relaxation-time argument above and on the SWMF methodology, not on measured evidence that it is better.
2. **Only smooth, low-Mach behaviour has been exercised.** The regime where the distinction would matter most — a strongly compressive, shock-forming event inside the partial-ionization zone — has been tested only in the ideal-gas shock tube, where γ is correct by construction. The

flare and shock scenarios run the non-release two-fluid solver, which this flux does not touch.

3. **The offset is upwinded by** $\mathrm{sign}(u_\star)$ **alone,** a piecewise-constant switch rather than a smooth function of the state. This is SWMF's construction and it is exact for a contact; no pathology was observed, because $u_\star \to 0$ also drives the energy flux to zero, but it is not smooth.

   Retired: the timestep inconsistency. Before the cutover the CFL was deliberately taken from the equilibrium Γ_1 speed so that the two comparison legs took identical steps, and this document previously recorded the resulting effective CFL as "about 0.62". That figure was wrong, and the inconsistency is now fixed; see section 12.2.

11.4 Gravity and reference-free hydrostatic balance

Active release path Face states use a common face gravitational potential on both sides, removing an artificial gravitational-energy jump from the numerical dissipation. Beyond that the release carries *no* well-balancing mechanism: the *interior* hydrodynamic operator is given no hydrostatic state in advance and subtracts no reference residual. This is a statement about the interior discretization only. The boundary closures still capture fixed reservoir data once from the initial condition — the outer back pressure `kOuterPRef` (section 9.4) and the lower reservoir state `kInnerTRef/kInnerPRes` (section 9.3) — as any truncated-domain problem must. What was removed is the dependence on a global hydrostatic *profile*, not every frozen quantity in the problem. Two classes of well-balanced scheme exist for the Euler equations with gravity — those that require the target equilibrium a priori [Berberich et al., 2019] and those that preserve a class of discrete equilibria to machine precision without knowing them [Käppeli and Mishra, 2014, Chandrashekar and Klingenberg, 2015]. **The release belongs to neither, and should not be described as a well-balanced scheme.** It removes the two discretization inconsistencies that dominated its hydrostatic imbalance and leaves a small, resolution-dependent residual, rather than constructing exact preservation of anything.

The defects were the missing predictor source of section 11.2.1 and the first-order lower ghost ladder of section 9.3. On the release mesh, measured as the numerical face mass flux of the initialized $u = 0$ atmosphere with no reference subtraction:

Configuration	$\max \lvert F^\rho \rvert$ at $t = 0$	20 s conduction-off $\max \lvert v \rvert$
Previous release	7.5×10^{-10}	$0.71\ \mathrm{m\,s^{-1}}$
+ predictor source	1.4×10^{-11}	$0.032\ \mathrm{m\,s^{-1}}$
+ trapezoidal lower ghosts	1.3×10^{-12}	$0.030\ \mathrm{m\,s^{-1}}$
float32 floor $\epsilon_{32}\rho(\lvert v \rvert + c_s)$	1.0×10^{-12}	—

The final defect is comparable to the estimated round-off scale of the stored state, and doubling the resolution does not reduce it (2.7×10^{-12} at $N = 1000$), so no clean truncation-error trend survives over the resolutions tested. This is deliberately weaker than identifying the residual *with* float32 representation: reconstruction, the EOS inversion, the nonuniform mesh and the boundary closures all contribute at this level, and the resolutions tested are too few to separate them. The 20 s residual velocity localizes in the upper transition region (2050–2120 km), not at the lower boundary, and is two to three orders of magnitude below the 15–40 $\mathrm{m\,s^{-1}}$ evaporation flow. A 5-step release-mesh regression asserting this reference-free behaviour is part of the test suite.

Retired The previous release stored the initialized equilibrium U^{eq}, evaluated its explicit residual once, and advanced

$$\frac{\mathrm{d}U}{\mathrm{d}t} = R(U) - R(U^{\mathrm{eq}}), \tag{11.21}$$

a δ-form equilibrium-reference method that made that one chosen state an exact discrete fixed point. It was removed for three reasons. It bound the solver to a pre-known global equilibrium. The residual was evaluated once, at $t = 0$ and at the first step's Δt, and then subtracted unchanged for the rest of the run — including after the transition region had completely restructured, and at every subsequent Δt, so the "exact fixed point" property held strictly only at $t = 0$. And, decisively, it no longer changed the answer: with the corrected discretization, enabling it altered the 100 s evaporation solution by 0.03% (0.15% at $N = 1000$) and the 1000 s solution by 0.4% in analysis-window velocity, 0.5% in peak velocity, and 0.9% in column mass drift. `Grid::eq_wb`, `eq_state` and `eq_residual` still exist for the two-fluid research solver, which never received the predictor source term; a release `Grid` sets `eq_wb=false` and the `ISO_EQ_WB` knob is gone.

Active release path The $t = 0$ face mass flux F^{ρ}_{ref} remains useful as a *diagnostic* baseline, and boundary diagnostics still report $F_{\text{eff}} = F^{\rho}_{\text{total}} - F^{\rho}_{\text{ref}}$ — the change in transport since initialization. It is no longer anything the solver subtracts. A smooth F_{eff} can coexist with oscillatory cell-centered ρv because the central and diffusive flux components compensate.

Chapter 12

Implicit Conduction and Time Integration

12.1 Nonlinear backward-Euler conduction

Active release path At fixed density, the conduction stage solves

$$e_{\text{int}}(\rho, T^{n+1}) - e_{\text{int}}(\rho, T^{*}) - \Delta t\, \nabla_s \cdot \left[K(T^{n+1}) \nabla_s T^{n+1}\right] = 0, \tag{12.1}$$

where

$$K = \kappa_e + \kappa_n, \tag{12.2}$$

and T^* is the temperature decoded from the post-hydrodynamic state U^*. Conductivity, Saha composition, and effective heat capacity are refreshed each nonlinear iteration.

The quasi-Newton matrix uses C_V^{eq} on the diagonal and treats the updated face conductivities in a Picard-like manner. Each iteration solves a tridiagonal system with the Thomas algorithm. Convergence is accepted only after evaluating the full nonlinear residual at the updated temperature. A small update that fails the residual test is reported as stagnation; reaching the iteration limit is an error.

On refined meshes, interior physical conductivity uses the width-weighted series resistance of the two half cells. Uniform meshes retain the historical arithmetic average. The production numerical-diffusivity multiplier is zero, so no artificial conductivity enters this solve. Closed-boundary tests verify that the weighted total-energy contribution telescopes on irregular grids.

12.2 Timestep control

Active release path The hydrodynamic timestep is

$$\Delta t_{\text{ac}} = C_{\text{CFL}} \min_i \frac{\Delta s_i}{|v_i| + c_i}, \qquad c_i^2 = \frac{\gamma_{\text{sig},i}\, p_i}{\rho_i}, \tag{12.3}$$

where γ_{sig} is the acoustic index of the *numerical flux in force*, not unconditionally the equilibrium one:

$$\gamma_{\text{sig}} = \begin{cases} \max(\Gamma_1,\ 5/3)\,, & \text{release Godunov flux (section 11.3.2)}, \\ \Gamma_1, & \text{Roe and Rusanov reference fluxes.} \end{cases} \tag{12.4}$$

The rule exists so that C_{CFL} always means the CFL of the scheme actually running. The release flux propagates the frozen waves of section 11.3.3, which in the partial-ionization interior travel about 1.24 times faster than the equilibrium waves; sizing the release step from Γ_1 would leave that factor as an undeclared margin. The max form, rather than a bare constant, keeps the bound conservative should the table ever return an index above the monatomic value, and it keeps the Roe comparison mode internally consistent rather than merely unchanged. Optional cooling, heating, or electron-energy physics can add smaller limits in non-release configurations. In the validated reference calculation, every step was acoustic limited.

Active release path **Measured effect on the canonical release configuration: none, and the reason is worth recording.** The CFL-limiting cell of the $N = 500$/R4 column is the topmost cell (2153 km, $\sim 22\,\mathrm{kK}$, hydrogen fully ionized, $\Delta s = 270\,\mathrm{m}$), where the tabulated Γ_1 is already 5/3. The large frozen/equilibrium speed gap lives in the partial-ionization interior, whose sound speed is far lower and which is nowhere near the CFL limit. Measured on both the $t = 0$ and the $t = 100\,\mathrm{s}$ states, the two rules select the same cell, give Δt ratios of $1 - 10^{-6}$, and produce the identical step count (17781 over 100 s) and end time. Measured against the frozen wave speeds, the *equilibrium*-sized step gave a maximum effective CFL of exactly 0.500. An earlier version of this document reported an effective CFL of about 0.62 for this flux; that figure assumed the limiting cell carried $\Gamma_1 \approx 1.09$ and is withdrawn.

The change is nonetheless required, as a guarantee rather than a correction: on a different mesh, a deeper domain, a different outer temperature, or a refinement that places a partial-ionization cell at the CFL limit, only Eq. (12.4) bounds the release scheme correctly.

The executable fallback is $C_{\mathrm{CFL}} = 0.25$. The low-I/O production launcher supplies 0.50, which has been validated for the canonical coarse-equivalent $N = 500$, R4, 661-cell physical-conduction-only configuration on the measured Apple Silicon system. That sweep was run under the Roe flux and the equilibrium signal speed; it transfers to the release flux because Δt and the step count are unchanged on this exact configuration, as measured above, and for no more general reason. It is not a universal stability guarantee.

12.3 Active release update path

Active release path One step on the three-variable state of Eq. (10.1) — explicit hydrodynamics Lie-split from implicit conduction — is

$$\boxed{U^n \xrightarrow{\text{explicit MUSCL–Hancock/Godunov hydrodynamics}} U^* \xrightarrow{\text{implicit physical conduction}} U^{n+1}.}$$

More explicitly:

1. decode U^n once through the equilibrium closure, caching $(T, x, p, \Gamma_1, e_{\mathrm{int}})$ per cell and for the two ghost layers;
2. reconstruct $(\log \rho, v, \log p)$ to cell i's own two faces with the MC3/Koren limiter, and take the Hancock half step of Eq. (11.3), which applies the same momentum source as the corrector; decode the half state through the caloric EOS;
3. rebuild all four face reconstructions from the time average $\frac{1}{2}(W^n + W^p)$, close them into equilibrium face states, and form the SWMF exact-Riemann Godunov fluxes of section 11.3.2, using face-local Rusanov only if the exact solve or its sample is inadmissible;
4. evaluate the explicit hydrodynamic RHS — flux divergence plus momentum source, with no reference subtraction — and advance density, momentum, and energy to the intermediate state U^*;

5. solve the nonlinear backward-Euler physical-conduction problem at fixed ρ and m, writing back $E^{n+1} = e_{\mathrm{int}}^{n+1} + m^2/2\rho + \rho\Phi$.

The two stages are composed by *first-order Lie splitting*: the conduction solve takes its initial temperature from the completed hydrodynamic state U^*, and the composition is not symmetrized about it. The MUSCL–Hancock construction supplies second-order space and time centering for the *homogeneous* flux evolution, but the corrector evaluates the momentum source at U^n rather than at the half-time state; in the source-only limit $\mathrm{d}U/\mathrm{d}t = S(U)$ the update degenerates to forward Euler, so the complete sourced hydrodynamic stage is not formally second order in time either. Composed with backward-Euler conduction, the step is first order in time overall. **No second-order temporal accuracy is claimed for the release, for the hydrodynamic stage or for the composition.** Raising the order of the source quadrature and symmetrizing the splitting are two separate questions, neither addressed here. This is acceptable here because the timestep is set by the acoustic CFL condition while the physics of interest evolves at $|v| \ll c_s$, so the splitting error per step is far below the acoustic scale that sets Δt; whether a Strang composition would measurably improve the long-duration evaporation solution is a separate numerical question that has not been tested and is not addressed here. There is no projection, repacking, or manifold-repair stage between the two: U is already the physical state, and Saha equilibrium enters only through the EOS closure that decodes it. Those two stages are the entire release timestep. The release performs no finite-rate ionization source step, and artificial conduction, TRAC, beam or coronal heating, radiative cooling, drag, separate thermal equilibration, and floor/slaving stages are not merely disabled here — they are not implemented in this solver at all, so no default-off flag stands between the reader and the algorithm. They belong to the two-fluid research solver described in the next section. For the same reason the release timestep control has only one constraint, the acoustic CFL condition: there is no volumetric source term to impose a heating-timescale cap, and the implicit conduction solve is unconditionally stable.

12.4 Legacy/full two-fluid update path

Implemented but inactive The older, non-release sequence is

$$\boxed{\begin{array}{c}\text{hydrodynamics} \to \text{drag} \to \text{thermal equilibration}\\ \to \text{conduction} \to \text{ionization} \to \text{heating/cooling}\\ \to \text{floors/slaving}\end{array}}$$

It operates on the independent seven-row ion, neutral, and electron state and uses fixed-γ pressure updates. Since the release was decoupled it is a genuinely separate solver: it owns its own conserved-state width, its own packing convention, its own boundary construction, and its own source stages, and its entry points reject a Grid that carries a Γ_1 table. It remains valuable for two-fluid drift, three-temperature, and finite-rate-ionization experiments (`model_gentle`, `model_c7`, `model_flare`, `analytic_canopy`, `pfss_field_line`), but it must not be used to describe the active release algorithm.

Chapter 13

Static Refinement and Inactive Artificial Diffusion

13.1 Mesh construction

Active release path Refinement is static, not adaptive mesh refinement. The face grid is built once before the initial state, and no regridding, remapping, or runtime adaptation occurs.

`ISO_NS` specifies the coarse-equivalent cell count. Refinement only adds cells. Two profiles exist:

lower a finite refined band followed by a geometric grade back to the coarse spacing;

outer a coarse region, a geometric coarse-to-fine grade, and a fine band extending to the outer face.

The adjacent cell-width ratio is capped at 1.1. The requested transition width is a minimum; it is widened when necessary to satisfy the ratio cap. Endpoints and the fully refined-region boundary are exact faces.

For the release outer profile, coarse-equivalent $N = 500$ produces 661 actual cells: coarse spacing about 1.105 km and fine spacing about 0.276 km.

13.2 Historical local numerical-conduction scaling

Historical The original implementation stored one global diffusivity proportional to the coarse spacing. Local refinement then reduced the wall stencil spacing without reducing the artificial conductivity, increasing the total conductive drive. A later face-local formulation stored C_{num} and evaluated $\chi_{\mathrm{num},f} = C_{\mathrm{num}}\Delta s_f$ independently at every face. A factor-four refined band therefore received approximately one quarter of the coarse-region diffusivity.

That change removed the refinement-induced increase in evaporation flux and preserved the uniform-grid result exactly, making it a useful controlled diagnostic. It has since been superseded for production: the artificial term was removed from the release conduction operator altogether, so the release advances physical conduction only by construction. The formulation survives only inside the two-fluid Stage-D operator, for historical and diagnostic comparisons, and is not release stabilization.

Part V

Implementation and Performance

Chapter 14

Code Architecture

14.1 Primary modules

File or module	Current responsibility
`scenarios/model_column.cpp`	Release-mode activation, domain and mesh parameters, PCHIP/Saha-HSE initial state, lower and upper boundaries, equilibrium reference. The same file also builds the shared two-fluid initial state for `model_gentle`, whose source controls are rejected in release mode.
`scenarios/model_c7.cpp/.hpp`	Model C7 tables, historical interpolator, gamma-mode PCHIP temperature, photoionization support for legacy kinetic studies.
`scenarios/mesh.cpp/.hpp`	Uniform and static refined face-grid construction, profile parsing, geometric grading, ratio-cap enforcement.
`src/eos.cpp`, `eos.hpp`	*Shared.* Saha closure, caloric EOS, derivatives, table parser/interpolator, temperature inversion, mixture decode and equilibrium face mapping.
`src/grid.cpp`, `chromosphere.hpp`	*Shared.* Grid: mesh geometry, prescribed $B(s)$, gravity, physical constants, metric caches, runtime flags, both state-row index sets, both ghost-buffer sets, diagnostic captures. Declares neither solver's entry points.
`src/single_fluid/mixture.hpp`	*Release solver.* Governing equations, software contract, and the complete release API.
`src/single_fluid/mixture.cpp`	*Release solver.* Three-variable state decode and packing, metric-aware $(\log\rho, v, \log p)$ MUSCL reconstruction, equilibrium face states, SWMF exact-Riemann Godunov flux with face-local Rusanov fallback (Roe and Rusanov reference solvers alongside), MUSCL–Hancock predictor/corrector with a source-carrying half step, geometric and gravitational momentum source, acoustic timestep.
`src/single_fluid/integrator.cpp`	*Release solver.* Nonlinear backward-Euler *physical* conduction and the two-stage release timestep.
`src/two_fluid/two_fluid.hpp`	*Two-fluid research solver.* Seven-row solver API: packed-state helpers, fluxes, source stages, integrators.

`src/two_fluid/state.cpp`	*Two-fluid research solver.* Packed seven-row indexing, ghost-layer access, `cons2prim/prim2cons`, limiter algebra.
`src/two_fluid/flux.cpp`	*Two-fluid research solver.* Fixed-γ physical fluxes, spectral radii, and explicit sources.
`src/two_fluid/rhs.cpp`	*Two-fluid research solver.* Seven-row reconstruction, predictor/corrector hydro operator, drag and conduction RHS.
`src/two_fluid/integrators.cpp`	*Two-fluid research solver.* Timestep limits, drag, thermal equilibration, conduction (including the historical artificial diffusivity), finite-rate ionization, TRAC, heating/cooling, floors, and the two-fluid update sequence.
`physics.hpp`	Inline formula library: collision frequencies, conductivities, radiative losses, heating rates, TRAC broadening, and the ionization/recombination rate coefficients. Used by the two-fluid solver and by EOS diagnostics; the release solver does not include it.
`chromo_main.cpp`	Command-line contract, fail-closed mode checks, run controls, output scheduling, diagnostics, termination reporting.
`parallel.hpp`	Static-schedule OpenMP loops and deterministic lowest-cell exception propagation.
`scripts/run_chromo_omp.sh`	Generic 12-thread OpenMP launcher.
`scripts/run_chromo_realtime.sh`	Validated low-I/O 12-thread launcher with CFL 0.50.
`tests/chromo_tests.cpp` and CTest	Unit, regression, process-level activation, table, output, and mesh validation.

14.2 Runtime contracts

Release mode is deliberately narrow, and it is selected by loading a Γ_1 table — no toggle chooses between the two solvers. The `model_column` scenario supplies the production table unconditionally, so the scenario name and the solver identity are the same fact: `model_column` is always the release solver, and the historical two-fluid column is the separate `model_gentle` scenario. The retired `model_isentropic` alias has been removed. The driver requires the semi-implicit `full` integrator and an explicit `no-ionization` command-line selection. Any legacy two-fluid setting supplied to a release run — `SINGLE_FLUID`, `ENABLE_TE`, `ISO_TWO_FLUID`, `ISO_IONIZATION`, `ISO_GAMMA`, `ISO_COOLING`, `ISO_TRAC`, `ISO_CORONA`, `ISO_CHEAT`, `ISO_QFLUX`, `ISO_TBOOST`, `ISO_NUMERICAL_DIFFUSIVITY_MULT` — is rejected outright rather than silently ignored, because none of them has meaning for a single-fluid common-temperature mixture advanced by hydro and physical conduction alone. Incompatible requests fail before the initial state is built.

Release state snapshots carry the three conserved rows (ρ, m, E); legacy two-fluid snapshots carry seven. The header line records the width. A companion `.gamma_diag` stream reports physical mixture density, velocity, temperature, Saha fraction, number densities, total pressure, Γ_1, and conductivity. Because release conduction is physical only, its `kappa_solver` column is identically `kappa_physical`; it is retained for format stability with the existing analysis scripts. The `.outercond` sidecar likewise now reports a single physical face conductivity and flux rather than a physical/artificial split. Optional `.faceflux` and `.outercond` files are default-off read-only diagnostics and do not feed the solver.

Chapter 15

Performance Engineering

15.1 Serial bottlenecks and method-preserving optimization

The first gamma-table implementation spent most of its time repeatedly decoding the same cell, solving caloric inversions, allocating temporary vectors, and rebuilding thermodynamic quantities already known to a caller. The major optimization stages preserved the numerical method:

Historical The timings in the following table used the superseded 2638-cell configuration and document the optimization history rather than current release throughput.

Stage	Material change	Measured effect
EOS/cache restructuring	Fast zero/one-update caloric inversion, decoded-cell caches, known-temperature packing, scratch reuse, fewer redundant table lookups.	About 4.25× relative to the early gamma-table implementation.
Serial production cleanup	Algebraic pressure-density inversion, fused EOS evaluation, fewer repeated logarithms and Γ_1 calls, reduced temporary work.	124.7 s to about 78.1 s for the same 20 s, 2638-cell case; about 1.60×.
Final serial follow-up	Additional cache and call-path cleanup without changing accepted states.	Median about 76.1 s, 2.67 ms per step.
OpenMP	Parallel per-cell decode, face construction, and a persistent conduction region.	16-thread median 36.35 s; 2.17× same-session speedup.
Thread-count selection	Compare 12 and 16 threads on Apple Silicon.	12 threads within about 1% of maximum throughput and selected as the default.

These numbers are hardware- and configuration-specific. They document engineering impact, not an algorithmic complexity guarantee.

15.2 Deterministic OpenMP

Active release path OpenMP is opt-in at CMake configuration. The runtime uses static scheduling, index-disjoint writes, fixed per-thread reduction slots, and deterministic exception selection by the lowest physical cell. The generic and production launchers set

`OMP_DYNAMIC=FALSE`, `OMP_MAX_ACTIVE_LEVELS=1`, `OMP_NUM_THREADS=12`.

Serial, one-thread OpenMP, and 12-thread OpenMP outputs were byte-identical in the validated checks.

The conduction team is capped at four threads because the remaining work alternates parallel cell/face evaluations with global barriers and a serial tridiagonal solve. The Thomas algorithm and nonlinear synchronization are the principal limits to further scaling. Replacing them would be a numerical-algorithm project, not a minor threading edit.

15.3 Long-run controls

Active release path `CHROMO_T_END` requests an exact physical end time. When it is set without `CHROMO_STEP_CAP`, the legacy time-multiplier step cap is removed. `CHROMO_FRAME_DT` schedules snapshots in physical time without cumulative drift; the initial and final states are written exactly once. Every run reports its termination cause, requested and final physical times, final step, and effective step cap.

The current $N = 500$, R4, 661-cell $\log p$, physical-conduction-only run (measured under the Roe flux) reached its requested 4000 s end time in 712314 steps with 12 OpenMP threads. Every timestep was acoustic limited, which is now the only limiter the release timestep can report. The nonlinear conduction solve averaged about 2.02 Newton updates and used at most four.

Part VI

Verification and Numerical Results

Chapter 16

Verification by Numerical Risk

16.1 Thermodynamic consistency

Active release path The runtime table contains 501 temperature values and 57 density values over

$$3.2 \times 10^3 \le T \le 10^8 \text{ K}, \qquad 10^{12} \le n_{\mathrm{H}} \le 10^{26} \text{ m}^{-3}.$$

It was generated as classical pure hydrogen with excitation, Fermi, and Coulomb corrections disabled. The parser rejects missing or unknown metadata, malformed dimensions, nonrectangular axes, nonpositive Γ_1, and unsupported schemas.

The validation program compares all 28557 production nodes with the independent C++ Saha and analytic caloric closure. Key maximum discrepancies are:

Check	Maximum discrepancy
C++ Saha vs. CRASH transition relative difference	1.74×10^{-3}
Analytic internal energy vs. CRASH relative difference	1.14×10^{-6}
Analytic effective heat capacity vs. CRASH relative difference	8.26×10^{-4}
Independent analytic Γ_1 vs. raw CRASH absolute difference	5.18×10^{-4}
Runtime table node vs. raw CRASH Γ_1	1.40×10^{-14}
Coarse bilinear interpolation vs. direct CRASH at all 28000 cell midpoints	5.37×10^{-3} absolute
Caloric inversion against direct CRASH energy	3.50×10^{-7} relative

The table approaches 5/3 in neutral and fully ionized limits. Continuity probes across every internal table grid line found a maximum jump of about 1.93×10^{-11}. Small adiabatic perturbations through 10%, 50%, and 90% ionization reproduced $c_s^2 = \Gamma_1 p/\rho$ within 5×10^{-4} relative.

16.2 State and closure semantics

Tests pack (ρ, v, T) into the release conserved state, decode it back, and verify that density, temperature, internal energy, pressure, the Saha fraction, and the derived carrier densities n_e, n_{HI}, ρ_i, ρ_n are all recovered from the three stored rows alone, including the identity $e_{\mathrm{int}} - \frac{3}{2}p = x n_{\mathrm{H}} \chi_{\mathrm{H}}$.

Non-physical states — negative density, an energy below the gravitational plus kinetic carry, or a density outside the Γ_1 table — are required to throw rather than be silently repaired. Separate tests assert that the legacy two-fluid entry points refuse a release Grid and that the release entry points refuse a Grid without a Γ_1 table.

16.3 Hydrostatic preservation and boundaries

The Saha-HSE initializer is checked against the trapezoidal logarithmic-pressure relation. All cells and four ghosts are decoded through the release EOS and checked for consistent pressure, composition, energy, and gravitational potential. Five reference-free release-mesh steps on the initialized column retain $\max|v| = 5.8 \times 10^{-3}\ \mathrm{m\,s^{-1}}$, $\max|\rho v| = 4.2 \times 10^{-13}$ and $\|\Delta U\|/\|U\| = 5.1 \times 10^{-8}$; the assertion is tied to the release mesh because, with no frozen reference, the bound is resolution dependent (on a $14\times$ coarser grid the same quantity is $0.76\ \mathrm{m\,s^{-1}}$). Refined-mesh tests confirm that HSE ghosts use the actual local boundary spacing.

The hydro-temperature-decoupled upper boundary is tested to ensure that the hydrodynamic ghosts carry the live top temperature while the conduction solver continues to use the fixed external wall. Diagnostics independently reconstruct the accepted outer conductive face coefficient and flux.

16.4 Conduction conservation and nonlinear convergence

Closed-domain uniform and irregular-grid problems verify conservation of the physically weighted energy $\sum_i E_i \Delta s_i / B_i$, and mass and momentum are required to be bit-preserved by the stage. The accepted packed state is checked against a separately recomputed full backward-Euler residual. A strong chromosphere-to-corona profile, which drives $\kappa_e \propto T^{5/2}$ through five decades, must converge without leaving the EOS table. A dedicated regression asserts that enabling every retired two-fluid stage flag on a release Grid reproduces the release step bit for bit, which is the executable statement that those stages are absent from the release timestep rather than merely disabled.

16.5 Irregular-grid consistency

Manufactured and regression tests cover metric caches, reconstruction gradients, pressure operators, acoustic CFL scaling, series-resistance conduction, HSE boundaries, local artificial diffusivity, and the uniform-grid compatibility branch. Mesh tests verify positive widths, exact endpoints, refinement-only cell addition, the 1.1 adjacent-width cap, and outer refinement extending through the last face.

16.6 Serial/OpenMP equivalence

The same release case has been run with a serial build, an OpenMP build restricted to one thread, repeated 12-thread runs, and the selected CFL. The main state and gamma diagnostic outputs were byte-identical after excluding the output path embedded in the header. This establishes deterministic execution for the tested path.

16.7 CFL and duration validation

CFL values 0.125, 0.25, 0.35, 0.40, 0.45, and 0.50 were compared on matched physical times. All candidates passed non-finite-state, EOS-domain, Newton-convergence, limiter, and termination gates. Thermodynamic and conserved-field errors at CFL 0.50 remained orders of magnitude inside the predefined tolerances relative to CFL 0.125, and medium-duration differences remained smaller than the already accepted 0.25-versus-0.125 gap.

A residual velocity field of only a few m s^{-1} does not converge monotonically as the timestep is reduced. The evidence is consistent with float32 cancellation around a hydrostatic balance. Consequently, m s^{-1}-scale velocities are below the trustworthy numerical floor even though the thermodynamic state is stable. This limitation applies to CFL 0.25 as well as 0.50.

Chapter 17

Development Experiments and Their Disposition

Each experiment below records the question, the controlled change, the result, and the final disposition. Metrics are limited to those needed to support the conclusion.

17.1 Constant-γ baselines

Question. Could hydrodynamics, boundaries, and conduction be developed before the equilibrium EOS was complete?

Controlled change. Use fixed $\gamma = 5/3$, then $\gamma = 1.05$ as a nearly isothermal surrogate.

Result. These modes enabled isolation of hydrostatic and boundary errors, and $\gamma = 1.05$ reduced excessive pressure response. Neither includes hydrogen ionization energy or the correct acoustic derivative.

Disposition. Historical Retained for regression and closure comparison; not a physical substitute for the release EOS.

17.2 Kinetic-ionization experiments

Question. Could finite-rate ionization and recombination support a dynamic chromosphere with independent fluids?

Controlled change. Activate collisional, photoionization, radiative, and three-body channels in the legacy source sequence.

Result. The experiments exposed strong energy-balance and upper-boundary sensitivity, including collapse when ionization cooling was not balanced by a coronal energy source. They also motivated robust trace-species and drift handling.

Disposition. Implemented but inactive Technically preserved; not the release closure. Non-equilibrium ionization remains scientifically important but requires a separately validated model.

17.3 Transition to Saha/Γ_1

Question. How can ionization energy and acoustic response be included without an inconsistent partial EOS conversion?

Controlled change. Stage the work through table validation, analytic caloric closure, safeguarded inversion, manifold-consistent face reconstruction, variable-EOS conduction/sources, and Saha-HSE boundaries.

Result. Every active operator now consumes the same equilibrium thermodynamics; incompatible legacy modes fail closed.

Disposition. Active release path Current physical core.

17.4 PCHIP correction

Question. Why did the initial conductive flux contain steps at C7 knots?

Controlled change. Replace piecewise-linear temperature interpolation with shape-preserving C^1 cubic Hermite interpolation, then reintegrate HSE.

Result. The initialization retained every C7 knot while removing derivative discontinuities and producing a smooth initial physical heat flux.

Disposition. Active release path Gamma-table initialization; fixed-gamma interpolation intentionally unchanged.

17.5 Lower split conduction-to-hydro artifact

Question. Was a small low-chromosphere mass-flux bump a physical evaporation signal?

Controlled change. Hold the well-balancing configuration of the time fixed (equilibrium-reference, then the release default) and refine only the lower 0–700 km band.

Result. The bump scaled approximately linearly with local cell width and extrapolated close to zero. It arose when the explicit hydro step differentiated a pressure field modified by the split conduction stage.

Disposition. Historical Numerical coupling artifact; local lower refinement or tighter hydro-conduction coupling is the remedy. It is distinct from the upper-TR release study.

17.6 Upper-boundary diagnosis

Question. Was the top-cell reversal caused by an over-specified hydrodynamic boundary?

Controlled change. Decouple hydro ghost temperature from the fixed conductive wall while preserving fixed back pressure and velocity cap.

Result. The last-cell reversal disappeared, but the 2130–2150 km ρv ripple remained. This separated a boundary over-specification problem from an interior truncation-error problem.

Disposition. Active release path Decoupled boundary retained; ripple investigated separately.

17.7 Face-flux diagnosis

Question. Did the ρv ripple violate discrete continuity?

Controlled change. Record the central, Rusanov-diffusive, total, reference, and effective face mass fluxes without changing the state.

Result. F_{eff} remained smooth. Central and diffusive components were strongly anti-correlated, while cell-centered ρv oscillated around the conserved flux.

Disposition. Use F_{eff}, not cell-centered ρv, for quantitative transport at the upper boundary.

17.8 Uniform resolution scan

Question. Does the visible upper-TR ripple converge?

Controlled change. Run uniform $N = 500, 1000, 2000$ meshes with the same EOS, wall, boundary, limiter, and well balancing.

Result. Ripple and cell-to-face mismatch decreased at approximately first order. The mean effective mass flux did not converge because the realized conductive input also changed with resolution.

Disposition. Ripple classified as spatial truncation error; evaporation magnitude remained provisional.

17.9 Static outer refinement

Question. Can the upper ripple be reduced without globally refining the whole column?

Controlled change. Add a factor-two and factor-four outer refined band at coarse-equivalent $N = 1000$.

Result. The normalized ripple amplitude fell from 0.1366 to 0.0514 to 0.0094 while the factor-four mesh used 1320 rather than 4000 cells. The grading interface introduced no localized artifact.

Disposition. Active release path Outer static refinement adopted. The first study's globally coarse-scaled artificial conductivity was superseded.

17.10 Face-local numerical conduction

Question. Why did refinement increase the evaporation rate in the first outer-refinement study?

Controlled change. Replace the global coarse-scaled artificial diffusivity with $\chi_{\mathrm{num},f} = C_{\mathrm{num}} \Delta s_f$.

Result. Uniform results were unchanged; the refined outer diffusivity became one quarter of the coarse value; the refinement-induced flux increase disappeared; the artificial fraction of the outer conductive input fell with resolution.

Disposition. Historical Retained for controlled legacy comparisons; superseded in production by physical conduction only.

17.11 Historical N=1000/2000/4000 convergence evidence

Historical This superseded local-scaling sequence used the old nonzero artificial-conduction coefficient and ghost-center thermal boundary. It remains useful for explaining the transition to local scaling, but it is not release evidence:

Metric at 500 s	N=1000	N=2000	N=4000
Actual cells	1320	2638	5275
Fine width [m]	138.02	69.10	34.55
Total outer conductive input [W m^{-2}]	1.4768	0.9808	0.7938
Artificial fraction	0.2308	0.1289	0.0685
Mean F_{eff} [kg m^{-2} s^{-1}]	1.0566×10^{-9}	6.1004×10^{-10}	6.2250×10^{-10}
Cell-to-face mismatch amplitude	0.0250	0.0101	0.00253

The mass flux changed by -42.3% and then $+2.04\%$; the sequence is nonmonotone and does not support an apparent order or Richardson limit. The difference between the last two values is 35.8 times smaller than the difference between the first two, which is evidence of approaching convergence. However, total conductive input still changed by -19.1% on the last refinement. The limiting evaporation rate therefore remains provisional.

The outer thermal layer was not a one-cell feature at the highest resolution: it was about 10 km thick and spanned roughly 300 fine cells. The maximum temperature gradient lay inside the domain rather than at the wall. In that retired implementation, the wall gradient and conductive input remained first-order dominated by the ghost-center Dirichlet stencil. The release now places the datum at the physical face and uses the half-cell distance in equation (9.4).

Part VII

Current Status and Roadmap

Chapter 18

Trust Boundaries

18.1 Scientifically trustworthy components

The following statements are supported by code-level consistency checks and controlled tests:

- the release state lies on the intended pure-hydrogen Saha equilibrium manifold;
- pressure, internal energy, ionization energy, effective heat capacity, and the equilibrium acoustic speed are mutually consistent (the release flux and CFL use the frozen speed of section 11.3.3, which is exact for the split of Eq. (11.20) and needs no table);
- the CRASH-derived Γ_1 table is parsed strictly and agrees with independent Saha derivatives within documented error;
- the Model C7/PCHIP temperature profile is reintegrated in the active hydrostatic EOS;
- the reference-free hydrodynamic operator preserves that state to within the float32 round-off floor of the stored state;
- nonlinear conduction satisfies its backward-Euler residual and conserves energy in closed tests;
- nonuniform reconstruction, conduction, CFL, and HSE operators use actual mesh metrics;
- the effective face mass flux is a valid conserved transport diagnostic;
- serial and OpenMP execution are identical for the validated path;
- the selected long-run controls terminate and report correctly.

18.2 Numerically validated but configuration-specific results

The 12-thread default and CFL 0.50 validation apply to the coarse-equivalent $N = 500$, R4, 661-cell, physical-conduction-only release shape on the tested Apple Silicon workstation. A different mesh, scenario, source set, or hardware should repeat the acceptance sweep.

18.3 Provisional scientific results

The release calculation develops a coherent upward effective mass flux that is nearly uniform with height across the upper-chromosphere diagnostic window. The $N = 500$ run (Roe flux) reaches 4000 s with low face-flux roughness, but this is not a formally converged evaporation rate because:

1. the matched $N = 1000$ calculation was stopped at 2080 s rather than completed to 4000 s;
2. long-duration $N = 500$ mass flux is therefore not established as grid-converged to better than 10%;
3. radiation is absent in the reference convergence calculation;
4. the upper atmosphere and corona are not resolved self-consistently.

No absolute evaporation rate from the current calculation should yet be presented as a converged physical prediction.

Chapter 19

Remaining Limitations and Priorities

19.1 Outer conductive boundary

The release discretization question has been resolved: $T_{\text{face}} = 22000$ K is imposed at the physical outer face and the top-cell distance is $\Delta s_{\text{top}}/2$. The remaining limitation is physical rather than geometric. A fixed external temperature and back pressure do not respond to the evolving column and cannot replace a self-consistent coronal energy balance. Future coupling should replace these prescribed reservoir data with exchanged characteristic and conductive information.

19.2 Radiative balance

The reference evaporation calculation includes conductive heating but no radiative cooling. This isolates the numerical problem but omits a leading chromospheric/TR sink. The implemented radiation and TRAC modules need a separate convergence campaign before quantitative energy partition or quasi-steady evaporation is claimed.

19.3 Equilibrium versus non-equilibrium ionization

Saha equilibrium is a controlled release closure, not a universal chromospheric approximation. Dynamic hydrogen ionization can lag temperature changes substantially [Carlsson and Stein, 2002, Leenaarts et al., 2007]. Future work must compare reaction time scales with hydrodynamic and conductive time scales and validate a non-equilibrium model without reintroducing inconsistent pressure or energy bookkeeping.

19.4 Per-step Saha re-equilibration and the frozen Riemann solve

Active release path This is, in the author's assessment, the largest unresolved physical-consistency question in the released model, and it was brought into focus rather than created by the numerical-flux cutover.

The release now treats the composition as *frozen* inside the local Riemann problem, for the relaxation-time reasons of section 11.3.3. But the equation of state is unchanged: every decode, every reconstruction and every ghost closure evaluates the instantaneous Saha fraction $x_{\text{Saha}}(n_{\text{H}}, T)$ of

Eq. (3.7). The composition is therefore reset to equilibrium at the end of every timestep. Schematically, one step is

$$\underbrace{\text{Saha-equilibrated state}}_{U^n} \longrightarrow \underbrace{\text{frozen-composition hydrodynamic step}}_{\text{exact Godunov flux}} \longrightarrow \underbrace{\text{Saha-equilibrated state}}_{U^{n+1}}. \tag{19.1}$$

These are two separate modelling choices, and the first does not imply the second. It is coherent to freeze the composition over the Riemann fan while relaxing it over the step, and it is coherent to do neither; what is not coherent is to justify the first by a long relaxation time and the second by an implicitly short one.

Read charitably, Eq. (19.1) is an *operator-split instantaneous-relaxation* approximation: hydrodynamics at frozen composition, followed by a relaxation substep taken in the limit $\tau_{\text{ion}} \to 0$. Operator splitting of a stiff relaxation term against transport is standard and, in the genuinely stiff limit, correct: if the ionization state truly equilibrated on a timescale short compared with the step, projecting onto the Saha manifold each step would be the right thing to do, and the frozen Riemann solve would then be a numerical device that happens not to matter because the relaxation immediately erases its composition error. The difficulty is that this is the opposite of the regime invoked to justify the frozen solve.

Three timescales must be kept distinct, and conflating any two of them produces a wrong argument.

Numerical timestep $\Delta t \approx 5.5 \times 10^{-3}\,\text{s}$ in the release configuration. **This is not the relevant comparison.** Saha equilibrium is not invalidated by taking small steps; a numerical timestep is not a physical timescale, and an argument of the form "$\tau_{\text{ion}} \gg \Delta t$, therefore Saha fails" is simply an error. The step size controls only the accuracy of whatever splitting is used.

Physical dynamical/compressive time the time over which a fluid element's thermodynamic state actually changes: of order the acoustic crossing time of the structure of interest, a shock-front crossing at the short end and $\sim 60\,\text{s}$ for the chromospheric dynamical timescale quoted by Leenaarts [2020].

Ionization/recombination relaxation time τ_{ion}, computed for chromospheric hydrogen by Carlsson and Stein [2002]: $\sim 10^3$–$10^5\,\text{s}$ through the mid-chromosphere, $\sim 10^2\,\text{s}$ at the transition-region base, and 10–$10^3\,\text{s}$ even inside shocks.

The physically meaningful comparison is the second against the third. On that comparison τ_{ion} exceeds the dynamical time by one to three orders of magnitude through most of the modelled column, which is precisely the statement that hydrogen ionization is *not* in instantaneous equilibrium. **?** make the magnitude of the resulting error concrete: in their simulations the equilibrium ionization degree varies by more than twenty orders of magnitude between shocked and unshocked gas, while the actual non-equilibrium degree stays nearly constant. Leenaarts et al. [2007] report the thermodynamic consequence — LTE ionization acts as an energy buffer that suppresses shock temperatures by several thousand kelvin — and Leenaarts [2020] describes the artificial "preferred temperature" plateaus (near 6, 10 and 22 kK) that an LTE equation of state imposes and that vanish under non-equilibrium ionization.

The honest statement of the release model is therefore: the equilibrium decode is a known approximation whose error is *unquantified in this code.* It is not defended here as physically accurate for rapidly varying flow; it is a closure that makes the three-variable conserved state determinate and cheap, it is applied consistently everywhere in the solver, and it is what every quantitative result in this document was computed with. The quasi-static evaporation problem the release actually reports is the most favourable case for it — the flow is smooth, subsonic and slowly varying, so the composition error relative to a finite-rate treatment is smallest — and the least favourable case, a strong compressive event inside the partial-ionization zone, is exactly the regime the release does not claim.

Planned A physically more complete model would evolve the ionization state rather than enforce it: carry x (or the ion and neutral populations) as an additional evolved variable with finite-rate collisional and radiative ionization/recombination source terms, so that the frozen Riemann solve and the composition update are consistent by construction and Eq. (19.1) is replaced by a genuine relaxation with a physical τ_{ion}. The non-release two-fluid solver already implements such a network (section 5.1), which makes this a matter of reconciling two existing formulations rather than of new physics. Two consequences should be anticipated before it is attempted: the conserved state widens and the caloric inversion is no longer a function of (ρ, e_{int}) alone, and the release's careful separation between stored and derived quantities — currently one of its strongest correctness properties — would have to be redesigned rather than extended. Nothing in the present cutover attempts it, and it should not be attempted as a side effect of a numerical-method change.

19.5 Unresolved corona and coupling

The truncated domain treats the corona as an external pressure and thermal reservoir. It does not evolve a coronal energy balance, loop geometry, wave heating, or global mass inventory. Planned AWSoM coupling should replace prescribed reservoir data with exchanged characteristic, conductive, mass, momentum, and energy information while retaining a well-posed number of incoming boundary conditions.

19.6 Numerical precision

The packed state remains float32. Thermodynamic root finding and table interpolation use double precision, but the residual velocity around a hydrostatic state shows a float-level noise floor. Studies requiring m s^{-1}-scale velocities should first evaluate a double-precision conserved state or a perturbation-variable formulation.

19.7 Recommended order of future work

1. complete a matched long-duration $N = 1000$ Roe/$\log p$, physical-conduction-only comparison;
2. add radiative balance and TRAC under the same local-refinement framework;
3. quantify equilibrium-ionization error against a controlled finite-rate calculation;
4. define a characteristic coupling contract for AWSoM/SWMF;
5. only then broaden geometry, heating, and multidimensional scope.

Part VIII

Appendices

Appendix A

General Two-Fluid Equations and Source Stages

A.1 Conservative variables

Historical A representative full state is

$$U = (\rho_i, \rho_n, \rho_i V, \rho_n U, E_i, E_n, E_e)^T. \tag{A.1}$$

For fixed γ, the historical row energies are

$$E_i = \frac{p_i}{\gamma - 1} + \frac{1}{2}\rho_i V^2 + \rho_i \Phi, \tag{A.2}$$

$$E_n = \frac{p_n}{\gamma - 1} + \frac{1}{2}\rho_n U^2 + \rho_n \Phi, \tag{A.3}$$

$$E_e = \frac{p_e}{\gamma - 1}. \tag{A.4}$$

The active release path replaces these caloric relations with the mapping in equation (6.3).

A.2 Backward-Euler drag stage

The legacy drag update uses the center-of-mass velocity v_{cm} and drift $w = V - U$. For a relaxation rate λ,

$$w^{n+1} = \frac{w^*}{1 + \lambda \Delta t}. \tag{A.5}$$

Total momentum is unchanged. The decrease in relative kinetic energy is deposited as heat. This stage is stable in the strong-coupling limit; the release solver has no such stage because it carries only one momentum.

A.3 Backward-Euler reaction stage

Writing the ion fraction as f, the local kinetic network produces polynomial backward-Euler equations because electron-impact and three-body channels depend on powers of the electron density. The implementation selects the physically admissible root in $[f_{\min}, 1 - f_{\min}]$, updates carrier masses conservatively, and accounts for ionization energy according to the reaction channel. This source stage belongs to the two-fluid solver and does not exist in the release.

A.4 Historical multidimensional formulation

Historical The repository documentation also preserves the parent two-fluid MHD system with induction, Lorentz forces, and divergence control. It motivated the conserved layout and field-aligned reduction, but no current release result is obtained by evolving that multidimensional system. It should be treated as architectural background rather than implemented release functionality.

Appendix B

EOS Table Construction and Format

B.1 Generation

The production Γ_1 table is generated by `util/eos/tabulate_gamma.f90`. The CRASH calculation is restricted to classical pure hydrogen:

- excitation disabled;
- Fermi-gas correction disabled;
- Coulomb correction disabled;
- statistical-weight convention aligned with the coefficient-one Saha relation;
- constants aligned with the C++ EOS.

The full validation output contains additional CRASH columns, while the tracked runtime table contains only $\log_{10} T$, $\log_{10} n_{\mathrm{H}}$, and Γ_1, plus machine-readable metadata.

B.2 Runtime policy

The loader requires an exact table identifier, axis names, ordering, dimensions, and physics metadata. Production interpolation is bilinear in log coordinates. Out-of-domain states throw; clamping exists only through an explicit debug option. A checksum test guards the tracked table, and the runtime diagnostic header records the actual loaded path and SHA-256 value.

Appendix C

Implementation Status Matrix

Element	Status	Current interpretation
Saha equilibrium	Active release path	Physical composition closure
CRASH Γ_1 table	Active release path	Acoustic derivative only
Analytic p, e_{int}, C_V	Active release path	Caloric EOS and conduction derivative
Seven-row carrier storage	Retired	Removed from the release; now exclusive to the non-release two-fluid solver
Independent ion-neutral drift	Implemented but inactive	Legacy two-fluid path
Finite-rate hydrogen chemistry	Implemented but inactive	Legacy path and controlled experiments
Common-temperature conduction	Active release path	Nonlinear backward Euler
Separate electron conduction equation	Experimental	Three-temperature path only
Optically thick/thin radiation	Implemented but inactive	Two-fluid solver only; not a release stage
TRAC	Implemented but inactive	Two-fluid solver only; not a release stage
Beam heating	Experimental	Two-fluid solver only; flare studies
Volumetric coronal heating	Experimental	Two-fluid solver only; resolved-corona studies
SWMF-style exact-Riemann Godunov flux	Active release path	Release numerical flux: exact ideal-gas Riemann solve at frozen $\gamma = 5/3$ with the ionization energy in an upwinded passive offset, face-local Rusanov fallback, per-run fallback tally

Mixture Roe characteristic flux	**Implemented but inactive**	Reference/comparison solver (`ISO_RIEMANN=roe-local`); the previous release flux; equilibrium Γ_1 linearization
Flux-consistent CFL signal speed	**Active release path**	$\gamma_{\mathrm{sig}} = \max(\Gamma_1, 5/3)$ under the release flux, Γ_1 under the reference fluxes
Static outer refinement	**Active release path**	Release mesh option
AMR/regridding	**Planned**	Not implemented
Face-local artificial diffusion	**Historical**	Removed from the release conduction operator; retained inside the two-fluid Stage-D operator for historical diagnostics
Fixed-pressure outer reservoir	**Active release path**	One incoming hydrodynamic condition
Hydro/conduction temperature decoupling	**Active release path**	Release upper boundary
Characteristic outer boundary	**Planned**	Needed for coupling/generalization
AWSoM coupling	**Planned**	Not yet implemented

Appendix D

Release-Relevant Environment Variables

This chapter lists only variables that a canonical `model_column` release run actually accepts. Variables belonging to the historical two-fluid research path — which `model_column` rejects outright rather than silently ignoring — are listed separately in Appendix E. The distinction is exact: a variable appears here if and only if the release solver reads it.

D.1 EOS, domain, and physics

Variable	Meaning
`GAMMA_TABLE`	Path to the strict Γ_1 table; selects the release solver. The `model_column` scenario supplies the production path unconditionally, so this only redirects it.
`ISO_H_BASE`	Absolute lower height in km.
`ISO_DH`	Domain length in km.
`ISO_NS`	Coarse-equivalent cell count; refinement may add cells.
`ISO_HEAT_FLUX`	Enables the conduction-driven stage.
`ISO_T_TOP`	Absolute external conductive temperature at the physical outer face in K.

D.2 Boundary and mesh

Variable	Meaning
`ISO_HYDRO_T_DECOUPLE`	Use live top-cell temperature in hydro ghosts while retaining the fixed physical-face conduction datum.
`ISO_INNER_T_NEUMANN`	Replace lower fixed-temperature conduction condition by zero flux.
`ISO_VCAP`	Outer velocity cap in local sound-speed units.
`ISO_TJUMP_A`, `ISO_TJUMP_B`	Conductive-wall and historical second-ghost temperature multipliers.

ISO_REFINE_PROFILE	`lower` or `outer`; overrides generic mesh alias.
ISO_REFINE_FACTOR	Fine-to-coarse resolution ratio; 1 disables refinement.
ISO_REFINE_S_LO_KM	Start of fully refined region measured from the inner face.
ISO_REFINE_S_HI_KM	End of lower-profile refined region; unused by outer profile.
ISO_REFINE_TRANSITION_KM	Minimum geometric-grade width.
GRID_REFINE_*	Generic aliases used when corresponding `ISO_REFINE_*` values are absent.
ISO_RELAX_TIME	Optional initial no-conduction relaxation interval.

D.3 Reference-only overrides and diagnostics

The release solver accepts the variables below, but none of them is a production setting. The first block selects a reference numerical method for controlled comparison against the release default; the second block only enables read-only sidecar output and does not alter the solution.

Variable	Meaning
ISO_RIEMANN	Reference flux override: `roe-local` selects the mixture Roe characteristic flux of section 11.3.1 — the equilibrium-Γ_1 linearization that was the previous release flux — and `rusanov` the local Lax–Friedrichs flux. The release default is the SWMF-style exact-Riemann Godunov flux of section 11.3.2; no variable is needed to select it. Either override also reverts the CFL signal speed to Γ_1 (section 12.2).
ISO_RECONSTRUCTION	Reference reconstruction override: `lnrho-v-lnt` restores the temperature-based form. The release default is $(\log\rho,\ v,\ \log p)$; no variable is needed to select it.
ISO_LIMITER	Diagnostic override: `mc3`, `minmod`, or `first`; not for production.
ISO_MC3_BETA	MC3 beta under the diagnostic limiter override.
CHROMO_FACE_FLUX_DIAG	Enable read-only face-flux sidecar.
CHROMO_OUTER_COND_DIAG	Enable accepted outer-conduction sidecar.
CHROMO_GAMMA_DIAG	Enable decoded EOS-aware sidecar.

D.4 Run and OpenMP controls

Variable	Meaning
CHROMO_CFL	Runtime CFL override.
CHROMO_T_END	Requested physical end time.
CHROMO_STEP_CAP	Explicit positive step cap.
CHROMO_FRAME_DT	Snapshot cadence in physical seconds.

CHROMO_OUTPUT	Enable/disable main snapshots.
CHROMO_PROFILE	Enable timing/profile summary.
OMP_NUM_THREADS	OpenMP team size; launchers default to 12.
OMP_DYNAMIC	Launchers set false.
OMP_MAX_ACTIVE_LEVELS	Launchers set 1.

Appendix E

Non-Release Two-Fluid Research Environment Variables

Every variable in this chapter configures the historical seven-row two-fluid research solver, reached through the separate `model_gentle` scenario (and the other two-fluid scenarios). None of them has meaning for the single-fluid release, whose timestep is the explicit hydrodynamic update followed by the implicit physical conduction solve and nothing else. A canonical `model_column` run therefore *rejects* the master switches below before the initial state is built rather than silently ignoring them; their dependent sub-parameters are inert without those switches. This documentation is retained because the two-fluid path remains implemented, tested, and used for the historical studies recorded in this document.

Variable	Meaning
`ISO_GAMMA`	Historical fixed-γ index of the two-fluid solver; $\gamma = 1.05$ was the nearly isothermal surrogate. Mutually exclusive with `GAMMA_TABLE`.
`ISO_TWO_FLUID`, `ISO_IONIZATION`	Independent ion/neutral fluids and finite-rate ionization.
`ISO_COOLING`	Optically thick/thin radiative sink.
`ISO_TRAC`, `ISO_TRAC_TCHROM`, `ISO_TRAC_TCMAXFRAC`	TRAC enable/disable and cutoff controls.
`ISO_CORONA`, `ISO_CORONA_TOP_KM`	Extend the domain into a resolved corona.
`ISO_QFLUX`, `ISO_QFLUX0`, `ISO_QFLUX_ENHANCE`, `ISO_QFLUX_T_ON`, `ISO_QFLUX_RAMP`	Imposed outer-flux boundary and ramp.
`ISO_CHEAT`, `ISO_CHEAT_*`	Volumetric coronal-heating configuration.
`ISO_TBOOST`, `ISO_TBOOST_TC`, `ISO_TBOOST_W`	One-time initial-condition coronal superheat and its selector parameters.
`ISO_NUMERICAL_DIFFUSIVITY_MULT`	Multiplier on the historical C_{num} of the two-fluid Stage-D conduction operator. The release conduction operator has no such term.
`SINGLE_FLUID`, `ENABLE_TE`	Legacy runtime solver settings of the two-fluid path; also rejected by a release run.

Appendix F

Canonical Run Configurations and Diagnostics

F.1 Validated low-I/O reference run

```
CHROMO_T_END=1000 \
scripts/run_chromo_realtime.sh \
  outputs/model_column/model_column_release_1000s.txt \
  full no-ionization model_column - 20.0 no-cooling
```

The scenario supplies the production Gamma table, 1600–2153 km domain, $N = 500$/R4 mesh with 661 actual cells, physical-face 22000 K boundary, physical conduction only, and release/$\log p$ numerics. The launcher supplies 12 threads, CFL 0.50, and low-I/O runtime defaults. Explicit environment assignments override these settings and define a new validation target.

F.2 Diagnostic output semantics

Main snapshot. The three packed release rows (ρ, m, E) plus geometry metadata; authoritative for restart/raw conserved state. A two-fluid snapshot carries seven rows instead, and the header line records the width.

Gamma sidecar. Decoded physical mixture quantities. `kappa_solver` is identically `kappa_physical` because release conduction is physical only; the column is kept for format stability.

Face-flux sidecar. Central, diffusive, and total mass fluxes at selected faces, from which the diagnostic $F_{\mathrm{eff}} = F^{\rho}_{\mathrm{total}} - F^{\rho}_{\mathrm{ref}}$ is formed against the $t = 0$ baseline. The mass flux is the `mix::RHO` continuity flux; the release state has no carrier rows to sum. Read-only.

Outer-conduction sidecar. Accepted final-Newton physical-face temperature, half-cell boundary distance, physical face conductivity, and the resulting face flux. There is no artificial-conduction column, because the operator has no such term. Read-only.

Console/profile log. Termination reason, timestep-limit counts, Newton/inversion counters, OpenMP team information, and region timing.

Appendix G

Test Inventory

The release-relevant test groups are:

- strict table metadata, axes, interpolation, limits, continuity, and checksum;
- Saha, caloric energy, effective heat capacity, temperature inversion, and acoustic derivatives;
- release state round trips, derived-carrier identities, and face-state identities;
- release activation, solver isolation, and rejection of incompatible scenarios, integrators, and every historical two-fluid knob;
- a bit-for-bit check that enabling the retired two-fluid stage flags on a release Grid changes nothing, and a `model_gentle` smoke test that the retained seven-row path still builds and advances;
- PCHIP nodal and derivative behavior;
- Saha-HSE integration and all four ghost states;
- hydro/conduction temperature decoupling;
- nonlinear conduction residual, conservation, and strong-profile convergence;
- uniform and irregular-grid reconstruction, CFL, HSE, and conduction;
- lower and outer static refinement, grading edge cases, and the two-fluid local numerical diffusivity;
- reference-free hydrostatic preservation on the release mesh, and evaporation regression windows;
- face-flux and outer-conduction diagnostic neutrality;
- the release SWMF exact Riemann solver against the five standard reference star states, its $x/t = 0$ sampling, the degenerate stationary-contact and uniform-flow cases, and a shock-tube check that the energy offset of equation (11.18) is inert to the hydrodynamics; and, on the production right-hand side, that the release flux is conservative, differs measurably from both reference solvers, and counts rather than hides a fallback;
- that the CFL rule of Eqs. (12.3)–(12.4) uses the frozen signal speed on the release path and the equilibrium speed under the Roe reference flux, that the release step never exceeds the reference step, that no cell exceeds the requested CFL measured against the frozen waves, and that the frozen/equilibrium speed gap is genuinely large somewhere in the column, so the rule is not a disguised no-op;
- output scheduling, step-cap validation, and termination classification;
- serial/OpenMP process-level and output equivalence checks.

The exact assertion count is intentionally omitted because it changes as coverage grows. Release claims should be tied to named risks and passing test groups rather than a stale aggregate number.

Appendix H

Chronological Development Ledger

Phase	Main question	Lasting outcome
Early project	Can a plasma-neutral chromosphere be evolved in a field-aligned code?	Seven-row two-fluid architecture, source stages, C7 scenarios, conduction and radiation experiments; retained as a separate non-release solver.
Boundary/HSE work	Why do nominally static columns drift or reverse at the boundaries?	Discrete HSE reservoirs, log reconstruction, fixed upper back pressure, two-ghost consistency; equilibrium-reference well balancing, later superseded by completing the MUSCL–Hancock source treatment and the trapezoidal lower ghost ladder.
Constant-γ studies	Can conduction-driven response be isolated before a complete EOS?	Fixed 5/3 and 1.05 baselines; useful diagnostics but inadequate ionization thermodynamics.
Gamma stages 0–2	Is a CRASH-derived acoustic table reproducible and Saha consistent?	Strict pure-H table generation, metadata, checksums, whole-grid validation.
Gamma stages 3–4	Can conserved energy be inverted consistently?	Analytic caloric EOS, safeguarded inverse, conservative common-fluid mapping.
Gamma stages 5–6	Can hydro faces remain on the equilibrium manifold?	Initial $\log \rho, v, \log T$ reconstruction, common Γ_1 wave speed, and fail-closed activation; later superseded by $\log p$ reconstruction and Roe dissipation, and that in turn by the frozen-composition Godunov flux.
Gamma stages 7–9	Can sources, conduction, IC, and boundaries use the same EOS?	Nonlinear mixture conduction, total-energy source updates, PCHIP/Saha-HSE initialization, release sign-off tests.
Upper-TR diagnosis	Is the top ripple a boundary instability or flux error?	Hydro/conduction T decoupling, smooth effective face flux, first-order truncation interpretation.
Refinement	Can the ripple be resolved efficiently?	Static outer profile with bounded grading and actual-cell accounting.

Artificial conduction	Why did refined evaporation change incorrectly?	Face-local mesh scaling isolated the error; production later removed artificial conduction entirely.
Release numerics	How can low-Mach face pressure and dissipation be controlled?	$(\log \rho, v, \log p)$ reconstruction, local mixture Roe flux, and face-local Rusanov fallback.
Release flux cutover	Which acoustic response should the face Riemann problem use?	Frozen composition ($\gamma = 5/3$) via the SWMF exact-Riemann Godunov flux, with the CFL signal speed made consistent with it.
Performance	Can the release run at useful wall time without changing the method?	EOS/cache optimization, deterministic OpenMP, 12-thread default, validated CFL 0.50 and long-run controls.
Current	Is the evaporation rate converged?	Stable $N = 500$ evolution; matched long-duration $N = 1000$ completion remains necessary for a sub-10% claim.

Appendix I

Retired Scenarios and Archived Formulations

I.1 Retired active assumptions

The following should not reappear as current release descriptions:

- analytic-isentrope initialization for `model_column`; the current IC is Model C7/PCHIP with Saha-HSE reintegration;
- independent finite-rate ionization in the release solver;
- a common fixed-γ pressure law in the release EOS;
- centered upper pressure extrapolation across the top cell;
- a live top-pressure anchor with no external back pressure;
- a single upper ghost temperature serving both hydro and conduction in the validated configuration;
- globally coarse-scaled artificial diffusivity on a refined mesh;
- interpretation of cell-centered ρv ripple as a discontinuous conserved mass flux;
- presentation of a last-pair near-flat mass flux as formal convergence.

I.2 Archived but useful formulations

The full two-fluid equations, kinetic rate network, three-temperature exchange, TRAC, chromospheric and coronal radiation, beam heating, coronal heating, loop geometries, and multidimensional MHD derivations remain valuable. They support future work and explain past decisions. Their equations and references are retained in this record, but their status labels must accompany any reuse.

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